

AKTU : Engineering Mathematics-1

Vector Calculus

Lec-1

Today's Target

- ✓ Introduction to Vectors



By Gulshan sir

Vector differentiation:

- Gradient, Curl and Divergence and their Physical interpretation,
- Directional derivatives.

Vector Integration:

- Line integral
- Surface integral
- Volume integral
- Gauss's Divergence theorem
- Green's theorem and Stoke's theorem (without proof) and their applications.

Vector: A physical quantity is said to be a vector quantity if it has

- 1) Magnitude
- 2) Direction
- 3) Obey vector law of addition

Examples : Displacement , velocity, acceleration, force etc.

Scalar : A physical quantity that has magnitude only is known as scalar.

Examples : Distance, time, speed area, volume etc.

Representation of Vector



$$\overrightarrow{AB} \text{ or } \vec{a}$$

A $\rightarrow$ initial point

B $\rightarrow$ Terminal point

Magnitude

$$|\overrightarrow{AB}| \text{ or } |\vec{a}| \text{ or } a$$

Unit Vector

The unit vector in the direction of a vector $\vec{a}$ is denoted by $\hat{a}$

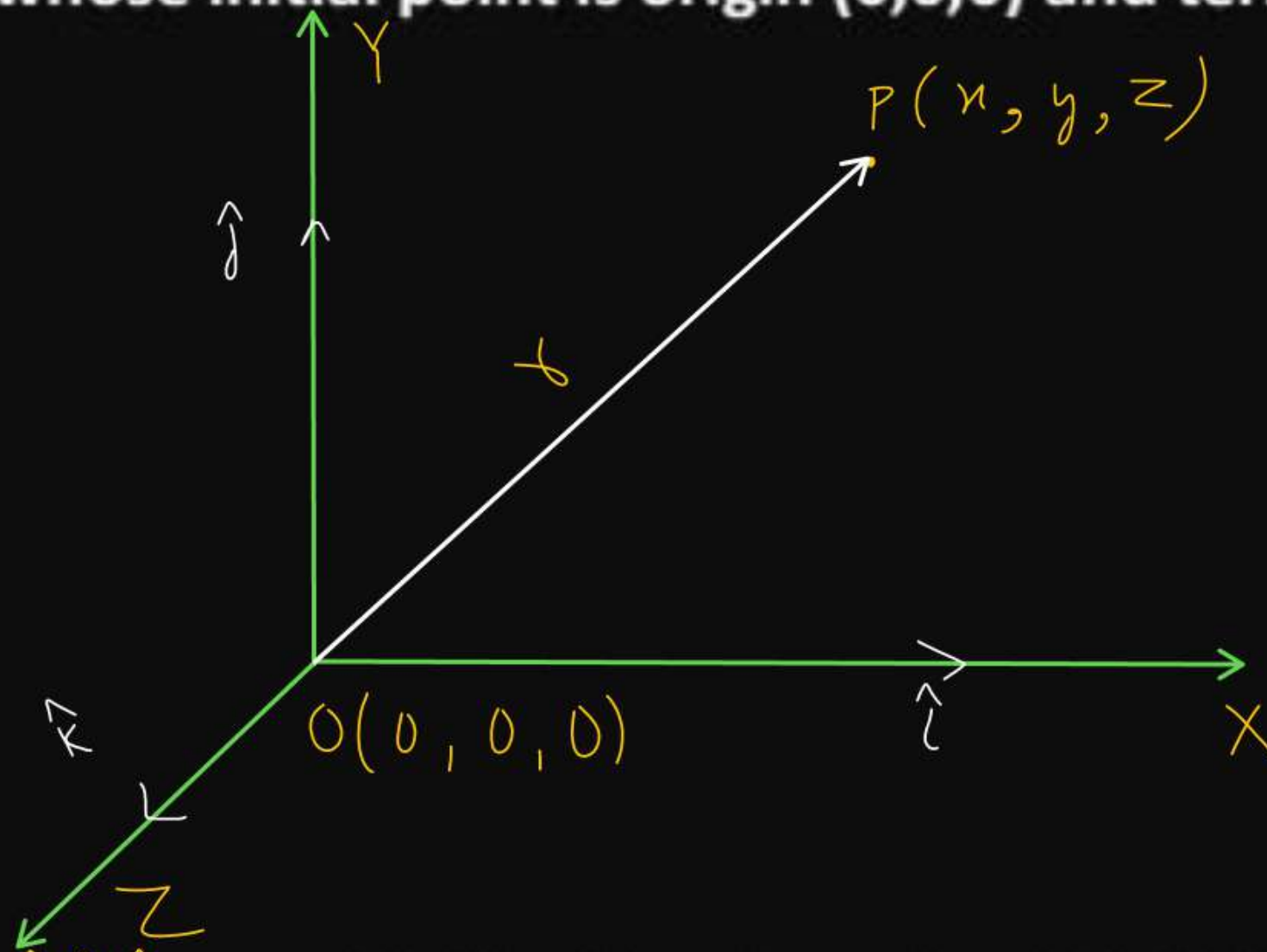
$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

Note : The magnitude of unit vector is unity

$$|\hat{a}| = 1$$

Position Vector

A vector whose initial point is origin (0,0,0) and terminal point is P(x,y,z)



Position vector of P

$$\vec{OP} = \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Magnitude

$$|\vec{OP}| = |\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$$

- Note :**
1. $\hat{i}, \hat{j}, \hat{k}$ are called unit vectors along OX, OY, OZ axes
 2. This form of vector is called component form of vector

a) x, y, z are called the scalar component of $\vec{r}$

b) $x\hat{i}, y\hat{j}, z\hat{k}$ are called the vector component of $\vec{r}$

Vector joining two points



$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$\vec{AB} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k} - (x_1\hat{i} + y_1\hat{j} + z_1\hat{k})$$

$$\vec{AB} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

$$|\vec{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Important Points : If $\vec{a} = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and $\vec{b} = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$

$$(i) \vec{a} + \vec{b} = (a_1 + a_2)\hat{i} + (b_1 + b_2)\hat{j} + (c_1 + c_2)\hat{k}$$

$$(ii) \vec{a} - \vec{b} = (a_1 - a_2)\hat{i} + (b_1 - b_2)\hat{j} + (c_1 - c_2)\hat{k}$$

$$(iii) \vec{a} = \vec{b} \text{ if and only if}$$

$$a_1 = a_2$$

$$b_1 = b_2$$

$$c_1 = c_2$$

Multiplication of vector by scalar

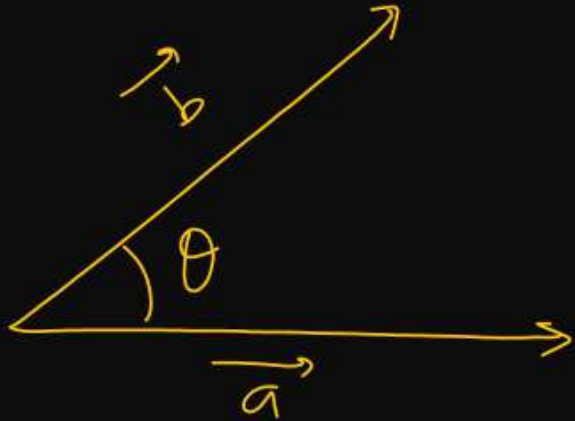
$$\text{Let } \vec{a} = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$$

$$\lambda\vec{a} = \lambda a_1\hat{i} + \lambda b_1\hat{j} + \lambda c_1\hat{k}$$

$$|\lambda\vec{a}| = \lambda|\vec{a}|$$

Scalar product (or DOT Product) of two Vectors

The scalar product of two non-zero vectors $\vec{a}$ and $\vec{b}$ is defined as



$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

Where θ is the angle between $\vec{a}$ and $\vec{b}$

$$0 \leq \theta \leq \pi$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

Results :

1. $\vec{a} \cdot \vec{b}$ is a real number
2. Two non-zero vectors $\vec{a}$ and $\vec{b}$ are perpendicular/orthogonal if and only if $\vec{a} \cdot \vec{b} = 0$

3. For unit vector $\hat{i}, \hat{j}, \hat{k}$

$$\hat{i} \cdot \hat{i} = 1$$

$$\hat{i} \cdot \hat{j} = 0$$

$$\hat{j} \cdot \hat{j} = 1$$

$$\hat{j} \cdot \hat{k} = 0$$

$$\hat{k} \cdot \hat{k} = 1$$

$$\hat{k} \cdot \hat{i} = 0$$

4. Scalar product in terms of component

$$\text{Let } \vec{a} = a_1 \hat{i} + b_1 \hat{j} + c_1 \hat{k}$$

$$\vec{b} = a_2 \hat{i} + b_2 \hat{j} + c_2 \hat{k}$$

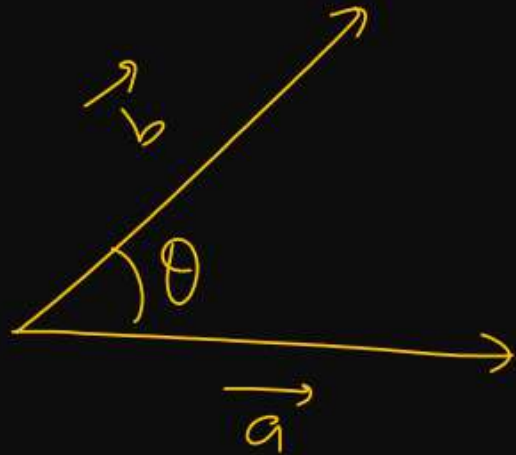
5.

$$\vec{a} \cdot \vec{b} = a_1 a_2 + b_1 b_2 + c_1 c_2$$

$$\textcircled{5} \quad \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

Vector product (or Cross Product) of two Vectors

The vector product of two non-zero vectors $\vec{a}$ and $\vec{b}$ is defined as



$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin\theta \hat{n}$$

Note : $\vec{a} \times \vec{b}$ is a vector perpendicular to both $\vec{a}$ and $\vec{b}$

Where

θ is the angle between $\vec{a}$ and $\vec{b}$

$$0 \leq \theta \leq \pi$$

$$\sin\theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}| \sin\theta}$$

Results :

(i) $\vec{a} \times \vec{b}$ is a vector

(ii) Two non-zero vectors $\vec{a}$ and $\vec{b}$ are parallel (or collinear) if and only if $\vec{a} \times \vec{b} = 0$

Note $\vec{a} \times \vec{a} = 0$
 $\vec{a} \times (-\vec{a}) = 0$

$$\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$$

(iii) For unit vectors

$$\hat{i} \times \hat{i} = 0$$

$$\hat{j} \times \hat{j} = 0$$

$$\hat{k} \times \hat{k} = 0$$

$$\hat{i} \times \hat{j} = \hat{k}$$

$$\hat{j} \times \hat{k} = \hat{i}$$

$$\hat{k} \times \hat{i} = \hat{j}$$

$$\hat{j} \times \hat{i} = -\hat{k}$$

$$\hat{k} \times \hat{j} = -\hat{i}$$

$$\hat{i} \times \hat{k} = -\hat{j}$$

(iv) Vectors product in terms of component

$$\text{Let } \vec{a} = a_1 \hat{i} + b_1 \hat{j} + c_1 \hat{k}$$

$$\vec{b} = a_2 \hat{i} + b_2 \hat{j} + c_2 \hat{k}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

Scalar triple product

$\vec{a}, \vec{b}, \vec{c}$

Important Point

(i) Scalar Triple product in terms of component

It represent the volume of parallelepiped

$$[\vec{a} \ \vec{b} \ \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b})$$

$$\text{Let } \vec{a} = a_1 \hat{i} + b_1 \hat{j} + c_1 \hat{k}$$

$$\vec{b} = a_2 \hat{i} + b_2 \hat{j} + c_2 \hat{k}$$

$$\vec{c} = a_3 \hat{i} + b_3 \hat{j} + c_3 \hat{k}$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

(ii) Three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar if

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0$$

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Vector Calculus

Lec-2

Today's Target

- ✓ Gradient
- ✓ Univ. Questions
- ✓ Practice Questions



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Vector Differentiation:

$\nabla \rightarrow Del$

or

Nabla Vector operator

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

Scalar function
(ϕ)

Vector function
(ϕ)

Gradient
 $\text{Grad } \phi = \nabla \phi$

(i) Divergence
 $\text{Div } \phi = \nabla \cdot \phi$

(ii) ~~curl~~ $\text{curl } \phi = \nabla \times \phi$

Gradient

The Gradient of scalar point function $\phi(x, y, z)$ is defined as

$$\text{grad}\phi = \nabla\phi$$

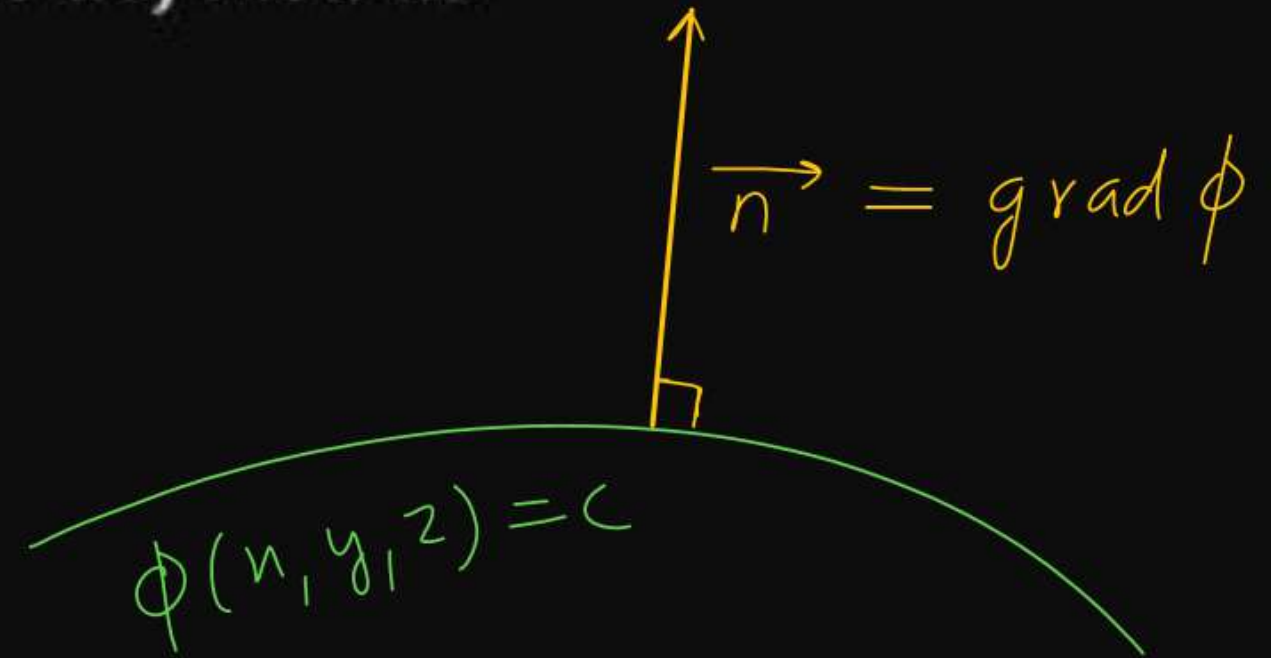
$$\text{grad}\phi = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}\right) \phi$$

$$\text{grad}\phi = \left(\hat{i} \frac{\partial\phi}{\partial x} + \hat{j} \frac{\partial\phi}{\partial y} + \hat{k} \frac{\partial\phi}{\partial z}\right)$$

Where $\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$ is called Delta or Nabla vector operator denoted by ∇

Geometrical Interpretation

$\nabla\phi$ is normal to the surface $\phi(x, y, z) = c$



Properties of gradient

(a) If ϕ is a constant scalar point function, then $\nabla\phi = \vec{0}$

(b) If ϕ_1 and ϕ_2 are two scalar point functions, then

(i) $\nabla(\phi_1 \pm \phi_2) = \nabla\phi_1 \pm \nabla\phi_2$

(ii) $\nabla(c_1\phi_1 \pm c_2\phi_2) = c_1\nabla\phi_1 \pm c_2\nabla\phi_2$ where c_1 and c_2 are constant

(iii) $\nabla(\phi_1\phi_2) = \phi_1\nabla\phi_2 + \phi_2\nabla\phi_1$

(iv) $\nabla\left(\frac{\phi_1}{\phi_2}\right) = \frac{\phi_2\nabla\phi_1 - \phi_1\nabla\phi_2}{\phi_2^2}$

Q.1 If $\phi = 3x^2y - y^3z^2$, find the grad ϕ at point $(2, 0, -2)$.

$$\phi = 3x^2y - y^3z^2$$

$$\text{grad } \phi = \nabla \phi$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \phi$$

$$= \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$= \hat{i} \frac{\partial}{\partial x} (3x^2y - y^3z^2) + \hat{j} \frac{\partial}{\partial y} (3x^2y - y^3z^2) + \hat{k} \frac{\partial}{\partial z} (3x^2y - y^3z^2)$$

$$= \hat{i} (6xy - 0) + \hat{j} (3x^2 - 3y^2z^2) + \hat{k} (0 - 2y^3z)$$

$$\text{At } (2, 0, -2)$$

$$\text{grad } \phi = \hat{i} (0 - 0) + \hat{j} (12 - 0) + \hat{k} (0 - 0)$$

$$\boxed{\text{grad } \phi = 12\hat{j}}$$

Practice Q. 1 If $\phi = 3x^2y - y^3z^2$, find the grad ϕ at point $(1, -2, -1)$

Ans. $-12\hat{i} - 9\hat{j} - 16\hat{k}$

Q.2 Find a unit vector normal to the surface $x^3 + y^3 + 3xyz = 3$ at the point $(1, 2, -1)$

$$x^3 + y^3 + 3xyz = 3$$

$$x^3 + y^3 + 3xyz - 3 = 0$$

$$\text{Let } \phi = x^3 + y^3 + 3xyz - 3$$

$$\text{grad } \phi = \nabla \phi$$

$$= \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$= \hat{i} \frac{\partial}{\partial x} (x^3 + y^3 + 3xyz - 3) + \hat{j} \frac{\partial}{\partial y} (x^3 + y^3 + 3xyz - 3) + \hat{k} \frac{\partial}{\partial z} (x^3 + y^3 + 3xyz - 3)$$

$$= \hat{i} (3x^2 + 3yz) + \hat{j} (3y^2 + 3xz) + \hat{k} (3xy)$$

At $(1, 2, -1)$

$$\text{grad } \phi = \hat{i} (3 - 6) + \hat{j} (12 - 3) + \hat{k} (6)$$

$$\text{grad } \phi = -3\hat{i} + 9\hat{j} + 6\hat{k}$$

Normal vector

$$\vec{n} = \text{grad } \phi$$

Practice Q.2 Find a unit vector normal to the surface $x^2y + 2xz = 4$
at the point $(2, -2, 3)$

Ans. $\frac{-\hat{i} + 2\hat{j} + \hat{k}}{3}$

Q.3 If $u = x + y + z$ $v = x^2 + y^2 + z^2$, $w = yz + zx + xy$
 prove that $\text{grad } u$, $\text{grad } v$ and $\text{grad } w$ are coplanar vectors.

$$u = x + y + z$$

$$v = x^2 + y^2 + z^2$$

$$w = yz + zx + xy$$

$$\text{grad } u = \nabla u$$

$$= \hat{i} \frac{\partial u}{\partial x} + \hat{j} \frac{\partial u}{\partial y} + \hat{k} \frac{\partial u}{\partial z}$$

$$= \hat{i} \times 1 + \hat{j} \times 1 + \hat{k} \times 1$$

$$\text{grad } u = \hat{i} + \hat{j} + \hat{k}$$

$$\text{grad } v = \nabla v$$

$$= \hat{i} \frac{\partial v}{\partial x} + \hat{j} \frac{\partial v}{\partial y} + \hat{k} \frac{\partial v}{\partial z}$$

$$\text{grad } v = 2x \hat{i} + 2y \hat{j} + 2z \hat{k}$$

$$\text{grad } w = \nabla w$$

$$= \hat{i} \frac{\partial w}{\partial x} + \hat{j} \frac{\partial w}{\partial y} + \hat{k} \frac{\partial w}{\partial z}$$

$$\text{grad } w = (z+y) \hat{i} + (z+x) \hat{j} + (x+y) \hat{k}$$

Scalar triple product

$$[\text{gradu} \text{ gradv} \text{ gradw}] = \begin{vmatrix} 1 & 1 & 1 \\ zn & zy & zz \\ y+z & n+z & n+y \end{vmatrix} = 2 \begin{vmatrix} 1 & 1 & 1 \\ n & y & z \\ y+z & n+z & n+y \end{vmatrix}$$

$$R_2 \rightarrow R_2 + R_3$$

$$= 2 \begin{vmatrix} 1 & 1 & 1 \\ n+y+z & n+y+z & n+y+z \\ y+z & n+z & n+y \end{vmatrix} = 2(n+y+z) \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ y+z & n+z & n+y \end{vmatrix}$$

Since R_1 and R_2 are same

$$[\text{gradu} \text{ gradv} \text{ gradw}] = 0$$

Hence gradu , gradv , gradw are
coplanar vectors

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Vector Calculus : Vector Differentiation

Lec-3

Today's Target

- ✓ Gradient **Part-2**
- ✓ Univ. Questions
- ✓ Practice Questions



By Gulshan sir

Gradient

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

$$\text{grad}\phi = \left(\hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \right)$$

Geometrical Interpretation

$\nabla\phi$ is normal to the surface $\phi(x, y, z) = c$

$$\vec{n} = \nabla\phi$$

Q.1 Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.

$$x^2 + y^2 + z^2 = 9$$

$$x^2 + y^2 + z^2 - 9 = 0$$

$$\text{Let } \phi_1 = x^2 + y^2 + z^2 - 9$$

$$z = x^2 + y^2 - 3$$

$$x^2 + y^2 - z - 3 = 0$$

$$\text{Let } \phi_2 = x^2 + y^2 - z - 3 = 0$$

Angle between ϕ_1 and $\phi_2 = \text{Angle between } \vec{n}_1 \text{ and } \vec{n}_2$

Where

$$\vec{n}_1 = \text{grad } \phi_1 = \nabla \phi_1$$

$$\vec{n}_1 = \left(\hat{i} \frac{\partial \phi_1}{\partial x} + \hat{j} \frac{\partial \phi_1}{\partial y} + \hat{k} \frac{\partial \phi_1}{\partial z} \right)$$

$$= \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2 - 9) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2 - 9) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2 - 9)$$

$$\vec{n}_1 = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

Q.2 Calculate the angle between the normals to the surface $xy = z^2$ at the points $(4, 1, 2)$ and $(3, 3, -3)$.

$$xy = z^2$$

$$xy - z^2 = 0$$

$$\text{Let } \phi = xy - z^2$$

$$\text{grad } \phi = \nabla \phi$$

$$= \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$= \hat{i} \frac{\partial}{\partial x}(xy - z^2) + \hat{j} \frac{\partial}{\partial y}(xy - z^2) + \hat{k} \frac{\partial}{\partial z}(xy - z^2)$$

$$= y\hat{i} + x\hat{j} - 2z\hat{k}$$

Normal vector at $(4, 1, 2)$

$$\vec{n}_1 = \text{grad } \phi$$

$$= y\hat{i} + x\hat{j} + z\hat{k}$$

$$\vec{n}_1 = \hat{i} + 4\hat{j} - 4\hat{k}$$

Normal vector at $(3, 3, -3)$

$$\vec{n}_2 = \text{grad } \phi$$

$$\vec{n}_2 = y\hat{i} + x\hat{j} + z\hat{k}$$

$$\vec{n}_2 = 3\hat{i} + 3\hat{j} + 6\hat{k}$$

Practice Q. 1 If θ is the acute angle between the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at the point $(1, -2, 1)$ show that $\cos\theta = \frac{3}{7\sqrt{6}}$

Q.3 Prove that $\nabla \log r = \frac{\vec{r}}{r^2}$

We know that

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$r^2 = x^2 + y^2 + z^2$$

$$r \frac{\partial r}{\partial x} = x$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

LHS

$$\nabla \log r = \hat{i} \frac{\partial}{\partial x} (\log r) + \hat{j} \frac{\partial}{\partial y} (\log r) + \hat{k} \frac{\partial}{\partial z} (\log r)$$

$$= \frac{1}{r} \times \frac{\partial r}{\partial x} \hat{i} + \frac{1}{r} \times \frac{\partial r}{\partial y} \hat{j} + \frac{1}{r} \times \frac{\partial r}{\partial z} \hat{k}$$

$$= \frac{1}{r} \times \frac{x}{r} \hat{i} + \frac{1}{r} \times \frac{y}{r} \hat{j} + \frac{1}{r} \times \frac{z}{r} \hat{k}$$

$$= \frac{1}{r^2} (x\hat{i} + y\hat{j} + z\hat{k})$$

$$\nabla \log r = \frac{\vec{r}}{r^2}$$

Hence proved

Q.4 Show that $\nabla r^n = nr^{n-2} \vec{r}$ and hence evaluate $\nabla \frac{1}{r}$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$r^2 = x^2 + y^2 + z^2$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

LHS

$$\nabla r^n = \hat{i} \frac{\partial(r^n)}{\partial x} + \hat{j} \frac{\partial(r^n)}{\partial y} + \hat{k} \frac{\partial(r^n)}{\partial z}$$

$$= nr^{n-1} \frac{\partial r}{\partial x} \hat{i} + nr^{n-1} \frac{\partial r}{\partial y} \hat{j} + nr^{n-1} \frac{\partial r}{\partial z} \hat{k}$$

$$= nr^{n-1} \times \frac{x}{r} \hat{i} + nr^{n-1} \times \frac{y}{r} \hat{j} + nr^{n-1} \times \frac{z}{r} \hat{k}$$

$$= \frac{nr^{n-1}}{r} (x\hat{i} + y\hat{j} + z\hat{k})$$

$$= nr^{n-2} \times \vec{r}$$

$$\boxed{\nabla r^n = nr^{n-2} \vec{r}}$$

Hence proved

Put $n = -1$

$$\nabla r^{-1} = -1 \times r^{-1-2} \vec{r}$$

$$\nabla \frac{1}{r} = -1 \times r^{-3} \times \vec{r}$$

$$\boxed{\nabla \frac{1}{r} = -\frac{\vec{r}}{r^3}}$$

Practice Q.2 If $\vec{r} = x\hat{j} + y\hat{i} + z\hat{k}$, then show that $\text{grad } r = \frac{\vec{r}}{r}$.

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AKTU : Engineering Mathematics-1

Vector Calculus : Vector Differentiation

Lec-4

Today's Target

- ✓ Directional Derivative
- ✓ Univ. Questions
- ✓ Practice Questions



By Gulshan sir

The directional derivative of a scalar point function f at a point $P(x, y, z)$ in the direction unit vector $\hat{a}$ is given by

$$\frac{df}{ds} = (\text{grad } f) \cdot \hat{a}$$

Note : The maximum/greatest rate of increase of $f = |\nabla f|$

Or

Maximum/Greatest value of directional derivative = $|\nabla f|$

Q.1 Find the directional derivative of $\phi(x, y, z) = x^2yz + 4xz^2$ at $(1, -2, 1)$ in the direction of $2\hat{i} - \hat{j} - 2\hat{k}$. Find also the greatest rate of increase of ϕ .

$$\phi(x, y, z) = x^2yz + 4xz^2$$

$$\nabla \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$\nabla \phi = \hat{i} \frac{\partial}{\partial x} (x^2yz + 4xz^2) + \hat{j} \frac{\partial}{\partial y} (x^2yz + 4xz^2) + \hat{k} \frac{\partial}{\partial z} (x^2yz + 4xz^2)$$

$$\nabla \phi = \hat{i} (2xyz + 4z^2) + \hat{j} (x^2z + 0) + \hat{k} (x^2y + 8xz)$$

$$\text{At } (1, -2, 1)$$

$$\nabla \phi = \hat{i} (-4 + 4) + \hat{j} (1 + 0) + \hat{k} (-2 + 8)$$

$$\nabla \phi = \hat{j} + 6\hat{k}$$

$$\text{Let } \vec{a} = 2\hat{i} - \hat{j} - 2\hat{k}$$

Unit vector

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

$$= \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{2^2 + (-1)^2 + (-2)^2}}$$

$$= \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{9}}$$

$$\hat{a} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}$$

$$\hat{a} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$$

Now

Directional derivative

$$\frac{d\phi}{ds} = (\text{grad } \phi) \cdot \hat{a}$$

$$= (0\hat{i} + \hat{j} + 6\hat{k}) \cdot \left(\frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k} \right)$$

$$= 0 - \frac{1}{3} + 6\left(-\frac{2}{3}\right)$$

$$= -\frac{1}{3} - 4$$

$$\frac{d\phi}{ds} = \frac{-1-12}{3}$$

$$\frac{d\phi}{ds} = -\frac{13}{3}$$

Greatest rate of increase

$$f \phi = |\nabla \phi|$$

$$= |\hat{j} + 6\hat{k}|$$

$$= \sqrt{1^2 + 6^2}$$

$$= \sqrt{37}$$

Q.2 Find the directional derivative of $\phi = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$ at the point $(3, 1, 2)$ in the direction of vector $yz\hat{i} + zx\hat{j} + xy\hat{k}$.

$$\phi = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$$

$$\nabla \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$\nabla \phi = \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-\frac{1}{2}} + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{-\frac{1}{2}} + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{-\frac{1}{2}}$$

$$\nabla \phi = \hat{i} \left[-\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{3}{2}} \cancel{x} \right] + \hat{j} \left[-\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{3}{2}} \cancel{y} \right] + \hat{k} \left[-\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{3}{2}} \cancel{z} \right]$$

$$\nabla \phi = -(x^2 + y^2 + z^2)^{-\frac{3}{2}} (x\hat{i} + y\hat{j} + z\hat{k})$$

Q.3 Find the directional derivative of the function $f = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of line PQ where Q is the point $(5, 0, 4)$.

in what direction it will be maximum? Find also the magnitude of this maximum.

$$f = x^2 - y^2 + 2z^2$$

$$\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

$$\nabla f = \hat{i} \frac{\partial}{\partial x} (x^2 - y^2 + 2z^2) + \hat{j} \frac{\partial}{\partial y} (x^2 - y^2 + 2z^2) + \hat{k} \frac{\partial}{\partial z} (x^2 - y^2 + 2z^2)$$

$$\nabla f = 2x\hat{i} - 2y\hat{j} + 4z\hat{k}$$

$$\text{At } (1, 2, 3)$$

$$\nabla f = 2\hat{i} - 4\hat{j} + 12\hat{k}$$

Now

$$P(1, 2, 3) \text{ and } Q(5, 0, 4)$$

$$\overrightarrow{PQ} = (5-1)\hat{i} + (0-2)\hat{j} + (4-3)\hat{k}$$

$$\overrightarrow{PQ} = 4\hat{i} - 2\hat{j} + \hat{k}$$

$$\overrightarrow{a} = 4\hat{i} - 2\hat{j} + \hat{k}$$

Q.4 Find the directional derivative of

$$\phi = 5x^2y - 5y^2z + \left(\frac{5}{2}\right)z^2x \text{ at the point } (1, 1, 1) \text{ in}$$

the direction of line $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$.

$$\phi = 5x^2y - 5y^2z + \frac{5}{2}z^2x$$

$$\nabla\phi = \hat{i}\frac{\partial\phi}{\partial x} + \hat{j}\frac{\partial\phi}{\partial y} + \hat{k}\frac{\partial\phi}{\partial z}$$

$$\nabla\phi = \hat{i}\frac{\partial}{\partial x}(5x^2y - 5y^2z + \frac{5}{2}z^2x) + \hat{j}\frac{\partial}{\partial y}(5x^2y - 5y^2z + \frac{5}{2}z^2x) + \hat{k}\frac{\partial}{\partial z}(5x^2y - 5y^2z + \frac{5}{2}z^2x)$$

$$\nabla\phi = \hat{i}(10xy + \frac{5}{2}z^2) + \hat{j}(5x^2 - 10yz) + \hat{k}(-5y^2 + 5xz)$$

$$\nabla\phi = \frac{25}{2}\hat{i} - 5\hat{j} + 0\hat{k}$$

Q.5 Find the directional derivative of the function $\phi = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction of the normal to the surface $x \log z - y^2 + 4 = 0$ at $(2, -1, 1)$.

$$\phi = xy^2 + yz^3$$

$$\nabla \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$\nabla \phi = \hat{i} \frac{\partial}{\partial x} (xy^2 + yz^3) + \hat{j} \frac{\partial}{\partial y} (xy^2 + yz^3) + \hat{k} \frac{\partial}{\partial z} (xy^2 + yz^3)$$

$$\nabla \phi = \hat{i} (y^2) + \hat{j} (2xy + z^3) + \hat{k} (3yz^2)$$

$$\text{At } (2, -1, 1)$$

$$\nabla \phi = \hat{i} + (-4+1)\hat{j} + \hat{k}(-3)$$

$$\nabla \phi = \hat{i} - 3\hat{j} - 3\hat{k}$$

$$\text{Let } f = x \log z - y^2 + 4$$

$$\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

Q.6 Find the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r}$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$r^2 = x^2 + y^2 + z^2$$

$$\cancel{r} \frac{\partial r}{\partial x} = \cancel{x}$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\text{Let } \phi = \frac{1}{r}$$

$$\nabla \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$= \hat{i} \frac{\partial}{\partial x} \left(\frac{1}{r} \right) + \hat{j} \frac{\partial}{\partial y} \left(\frac{1}{r} \right) + \hat{k} \frac{\partial}{\partial z} \left(\frac{1}{r} \right)$$

$$= \hat{i} \left(-\frac{1}{r^2} \right) \frac{\partial r}{\partial x} + \hat{j} \left(-\frac{1}{r^2} \right) \frac{\partial r}{\partial y} + \hat{k} \left(-\frac{1}{r^2} \right) \frac{\partial r}{\partial z}$$

$$= -\frac{1}{r^2} \left(\frac{x}{r} \hat{i} + \frac{y}{r} \hat{j} + \frac{z}{r} \hat{k} \right)$$

$$\nabla \phi = -\frac{1}{r^3} (x\hat{i} + y\hat{j} + z\hat{k})$$

Unit vector

$$\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}}$$

$$\hat{r} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r}$$

Directional Derivative

$$\frac{d\phi}{ds} = \nabla\phi \cdot \hat{\gamma} = \frac{-1}{r^3} (x\hat{i} + y\hat{j} + z\hat{k}) \cdot \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{r}$$

$$= \frac{-1}{r^4} (x^2 + y^2 + z^2)$$

$$= -\frac{1}{r^{4-2}} \cancel{r^2}$$

$$\frac{d\phi}{ds} = -\frac{1}{r^2}$$

Q.7 Find the directional derivative of $\frac{1}{r^2}$ in the direction of $\vec{r}$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

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AKTU : Engineering Mathematics-1

Vector Calculus : Vector Differentiation

Lec-5

Today's Target

- ✓ Divergence and Curl of a Vector Point Function
- ✓ Univ. Questions
- ✓ Practice Questions



By Gulshan sir

Divergence of a Vector Point Function

The divergence of a differentiable vector point function $\vec{V}$ is denoted by $\text{div } \vec{V}$ and is defined as

$$\text{div } \vec{V} = \nabla \cdot \vec{V}$$

where

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

$$\vec{V} = V_1 \hat{i} + V_2 \hat{j} + V_3 \hat{k}$$

$$\text{div } \vec{V} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (V_1 \hat{i} + V_2 \hat{j} + V_3 \hat{k})$$

$$\text{div } \vec{V} = \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}$$

Note : Divergence of a vector point function is a scalar point function.

Q.1 If $\vec{V} = xy^2\hat{i} + 2x^2yz\hat{j} - 3yz^2\hat{k}$, then find $\text{div } V$ at $(1, 1, 1)$.

$$\vec{V} = xy^2\hat{i} + 2x^2yz\hat{j} - 3yz^2\hat{k}$$

$$\text{div } \vec{V} = \nabla \cdot \vec{V}$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (xy^2\hat{i} + 2x^2yz\hat{j} - 3yz^2\hat{k})$$

$$= \frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(2x^2yz) - \frac{\partial}{\partial z}(3yz^2)$$

$$= y^2 + 2x^2z - 6yz$$

$$\text{At } (1, 1, 1)$$

$$\text{div } \vec{V} = y^2 + 2x^2z - 6yz$$

$$\text{div } \vec{V} = 1^2 + 2 \times 1 \times 1 - 6 \times 1 \times 1$$

$$= 1 + 2 - 6$$

$$= 3 - 6$$

$$\boxed{\text{div } \vec{V} = -3}$$

Q.2 If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, show that $\text{div } \vec{r} = 3$.

Univ. Exam

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\text{div } \vec{r} = \nabla \cdot \vec{r}$$

$$= \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)$$

$$= 1 + 1 + 1$$

$$\boxed{\text{div } \vec{r} = 3}$$

Solenoidal Vector :

If $\text{div } \vec{V} = 0$, then $\vec{V}$ is called solenoidal vector.

Univ. Exam

Q.3 Show that vector $\vec{V} = (x + 3y)\hat{i} + (y - 3z)\hat{j} + (x - 2z)\hat{k}$ is solenoidal.

$$\vec{V} = (x + 3y)\hat{i} + (y - 3z)\hat{j} + (x - 2z)\hat{k}$$

$$\text{div } \vec{V} = \nabla \cdot \vec{V}$$

$$= \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}$$

$$= \frac{\partial}{\partial x}(x + 3y) + \frac{\partial}{\partial y}(y - 3z) + \frac{\partial}{\partial z}(x - 2z)$$

$$\begin{aligned} \text{div } \vec{V} &= 1 + 1 + (-2) \\ &= 2 - 2 \\ &= 0 \end{aligned}$$

$$\boxed{\text{div } \vec{V} = 0}$$

Hence, $\vec{V}$ is a solenoidal vector

Curl of a Vector Point Function

The curl (or rotation) of a differentiable vector point function $\vec{V}$ is denoted by $\text{curl } \vec{V}$ and is defined as

$$\text{curl } \vec{V} = \nabla \times \vec{V}$$

where

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

$$\vec{V} = v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}$$

$$\text{curl } \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Note : Curl of a vector point function is also a vector point function.

Q.4 If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, show that $\text{curl } \vec{r} = \vec{0}$.

Univ. Exam

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\text{curl } \vec{r} = \nabla \times \vec{r}$$

$$\text{curl } \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}$$

$$= \hat{i} \left(\frac{\partial}{\partial y}(z) - \frac{\partial}{\partial z}(y) \right) - \hat{j} \left[\frac{\partial}{\partial x}(z) - \frac{\partial}{\partial z}(x) \right]$$

$$+ \hat{k} \left[\frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x) \right]$$

$$= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(0-0)$$

$$= \vec{0}$$

Irrotational Vector :

If $\text{curl } \vec{V} = \vec{0}$, then $\vec{V}$ is called irrotational vector.

Univ. Exam

Q.5 Show that vector field $\vec{V} = (\sin y + z)\hat{i} + (x \cos y - z)\hat{j} + (x - y)\hat{k}$ is irrotational.

$$\vec{V} = (\sin y + z)\hat{i} + (x \cos y - z)\hat{j} + (x - y)\hat{k}$$

$$\text{curl } \vec{V} = \nabla \times \vec{V}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin y + z & x \cos y - z & x - y \end{vmatrix}$$

$$\text{curl } \vec{V} = \hat{i} \left[\frac{\partial}{\partial y}(x - y) - \frac{\partial}{\partial z}(x \cos y - z) \right] - \hat{j} \left[\frac{\partial}{\partial x}(x - y) - \frac{\partial}{\partial z}(\sin y + z) \right]$$

$$+ \hat{k} \left[\frac{\partial}{\partial x}(x \cos y - z) - \frac{\partial}{\partial y}(\sin y + z) \right]$$

$$= \hat{i}(-1 + 1) - \hat{j}(1 - 1) + \hat{k}(\cos y - \cos y)$$

$$\boxed{\text{curl } \vec{V} = 0} \quad \text{Hence, } \vec{V} \text{ is irrotational}$$

Q.6 Find the divergence and curl of the vector

Univ. Exam

$$\vec{R} = (x^2 + yz)\hat{i} + (y^2 + zx)\hat{j} + (z^2 + xy)\hat{k}.$$

$$\text{curl } \vec{R} = 0$$

$$\text{div } \vec{R} = \nabla \cdot \vec{R}$$

$$= \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$= \frac{\partial}{\partial x}(x^2 + yz) + \frac{\partial}{\partial y}(y^2 + zx) + \frac{\partial}{\partial z}(z^2 + xy)$$

$$= 2x + 2y + 2z$$

$$\boxed{\text{div } \vec{R} = 2(x + y + z)}$$

$$\text{curl } \vec{R} = \nabla \times \vec{R}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + yz & y^2 + zx & z^2 + xy \end{vmatrix}$$

Hence $\vec{R}$ is an irrotational vector

$$= \hat{i} \left[\frac{\partial}{\partial y}(z^2 + xy) - \frac{\partial}{\partial z}(y^2 + zx) \right] - \hat{j} \left[\frac{\partial}{\partial x}(z^2 + xy) - \frac{\partial}{\partial z}(x^2 + yz) \right] + \hat{k} \left[\frac{\partial}{\partial x}(y^2 + zx) - \frac{\partial}{\partial y}(x^2 + yz) \right]$$

$$\text{curl } \vec{R} = \hat{i}(x - x) - \hat{j}(y - y) + \hat{k}(z - z)$$

Practice Q. 1 Find the $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ where $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$.

Univ. Exam

Ans $\text{div} = 6(x+y+z)$

$\text{Curl} = 0$

Practice Q. 2 Find the the value of K if the vector

$$\vec{V} = (x + ky)\hat{i} + (ky - 3z)\hat{j} + (x - 2z)\hat{k}$$

is a solenoidal vector

Univ. Exam

Ans. $K = 1$

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AKTU : Engineering Mathematics-1

Vector Calculus : Vector Differentiation

Lec-6

Today's Target

- ✓ Divergence and Curl (Part-2) ✓✓
- ✓ Univ. Questions
- ✓ Practice Questions



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- It gives the rate of flow originating per unit volume at any point in vector field.
- If the fluid is incompressible

$$\text{div } \vec{V} = 0$$

- If $\text{div } \vec{V} = 0$
 $\vec{V}$ is solenoidal vector function

Physical Interpretation of curl

- It gives angular velocity at any point in vector field.
- If $\text{curl } \vec{V} = 0$
 then $\vec{V}$ is irrotational vector

Note : for a constant vector $\vec{a}$

$$\text{div } \vec{a} = 0$$

$$\text{curl } \vec{a} = 0$$

$$\text{curl } \vec{a} = 0$$

Q.1 If $\vec{A} = xz^2\hat{i} + 2y\hat{j} - 3xz\hat{k}$ and $\vec{B} = 3xz\hat{i} + 2yz\hat{j} - z^2\hat{k}$, find the value of $\vec{A} \times (\nabla \times \vec{B})$

Univ. Exam

$$\nabla \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3xz & 2yz & -z^2 \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y}(-z^2) - \frac{\partial}{\partial z}(2yz) \right] - \hat{j} \left[\frac{\partial}{\partial x}(-z^2) - \frac{\partial}{\partial z}(3xz) \right] + \hat{k} \left[\frac{\partial}{\partial x}(2yz) - \frac{\partial}{\partial y}(3xz) \right]$$

$$= \hat{i}(0 - 2y) - \hat{j}(0 - 3x) + \hat{k}(0 - 0)$$

$$\nabla \times \vec{B} = -2y\hat{i} + 3x\hat{j}$$

Velocity Potential or Scalar potential

If vector $\vec{V}$ is an irrotational vector then

$$\text{curl } \vec{V} = 0$$

Let ϕ is a scalar potential.

$$\text{Let } \vec{V} = \nabla \phi$$

$$\vec{V} \cdot d\vec{r} = \nabla \phi \cdot d\vec{r}$$

$$\vec{V} \cdot d\vec{r} = \left(\hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \right) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k})$$

$$\vec{V} \cdot d\vec{r} = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$\vec{V} \cdot d\vec{r} = d\phi$$

$$d\phi = \vec{V} \cdot d\vec{r}$$

$$\int d\phi = \int \vec{V} \cdot d\vec{r}$$

$$\phi = \int \vec{V} \cdot d\vec{r}$$

Q.2 A fluid motion is given by $\vec{V} = (y + z)\hat{i} + (z + x)\hat{j} + (x + y)\hat{k}$.

(i) Is this motion irrotational? If so, find the velocity potential.

(ii) Is the motion possible for an incompressible fluid?

$$\vec{V} = (y + z)\hat{i} + (z + x)\hat{j} + (x + y)\hat{k}$$

The motion is irrotational if

$$\text{curl } \vec{V} = 0$$

$$\text{curl } \vec{V} = \nabla \times \vec{V}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y+z & z+x & x+y \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y}(x+y) - \frac{\partial}{\partial z}(y+z) \right] - \hat{j} \left[\frac{\partial}{\partial x}(x+y) - \frac{\partial}{\partial z}(y+z) \right] + \hat{k} \left[\frac{\partial}{\partial x}(z+x) - \frac{\partial}{\partial y}(y+z) \right]$$

$$= \hat{i}(x-x) - \hat{j}(x-x) + \hat{k}(x-x)$$

$$= 0\hat{i} - 0\hat{j} + 0\hat{k}$$

$$= \vec{0}$$

Hence, motion is irrotational

We know that

$$d\phi = \vec{v} \cdot d\vec{r}$$

$$d\phi = [(y+z)\hat{i} + (z+n)\hat{j} + (n+y)\hat{k}] \cdot [dn\hat{i} + dy\hat{j} + dz\hat{k}]$$

$$d\phi = (y+z)dn + (z+n)dy + (n+y)dz$$

$$d\phi = ydn + zdn + zdy + ndy + ndz + ydz$$

$$= ndy + ydn + ndz + zdn + ydz + zdy$$

$$d\phi = d(ny) + d(nz) + d(yz)$$

Integrate both side

$$\phi = ny + nz + yz + C$$

Motion is possible for incompressible fluid if

$$\text{div } \vec{v} = 0$$

$$\text{div } \vec{v} = \nabla \cdot \vec{v}$$

$$\text{div } \vec{v} = \left(\hat{i} \frac{\partial}{\partial n} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot [(y+z)\hat{i} + (z+n)\hat{j} + (n+y)\hat{k}]$$

$$\text{div } \vec{v} = \frac{\partial}{\partial n}(y+z) + \frac{\partial}{\partial y}(z+n) + \frac{\partial}{\partial z}(n+y)$$

$$\text{div } \vec{v} = 0 + 0 + 0 = 0$$

Q.3 A fluid motion is given by $\vec{v} = (y \sin z - \sin x)\hat{i} + (x \sin z + 2yz)\hat{j} + (xy \cos z + y^2)\hat{k}$.

Univ. Exam

Is the motion irrotational? If so, find the velocity potential.

We know that

$$d\phi = \vec{v} \cdot d\vec{r}$$

$$d\phi = [(y \sin z - \sin x)\hat{i} + (x \sin z + 2yz)\hat{j} + (xy \cos z + y^2)\hat{k}] \cdot [dx\hat{i} + dy\hat{j} + dz\hat{k}]$$

$$d\phi = (y \sin z - \sin x)dx + (x \sin z + 2yz)dy + (xy \cos z + y^2)dz$$

$$d\phi = y \sin z dx - \sin x dx + x \sin z dy + 2yz dy + xy \cos z dz + y^2 dz$$

$$d\phi = -\sin x dx + (y \sin z dx + x \sin z dy + xy \cos z dz) + y^2 dz + 2yz dy$$

$$d\phi = d(\cos x) + d(xy \sin z) + d(y^2 z)$$

$$\phi = \cos x + xy \sin z + y^2 z + C$$

Integrate both side

Q.4 Find the constants a, b, c so that

$$\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k} \text{ is irrotational}$$

If $\vec{F} = \text{grad}\phi$, show that $\phi = \frac{x^2}{2} - \frac{3y^2}{2} + z^2 + 2xy + 4xz - yz$.

Since $\vec{F}$ is an irrotational

$$\text{curl } \vec{F} = 0$$

$$\nabla \times \vec{F} = 0$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+2y+az & bx-3y-z & 4x+cy+2z \end{vmatrix} = 0$$

$$\begin{aligned} & \hat{i} \left[\frac{\partial}{\partial y} (4x+cy+2z) - \frac{\partial}{\partial z} (bx-3y-z) \right] \\ & - \hat{j} \left[\frac{\partial}{\partial x} (4x+cy+2z) - \frac{\partial}{\partial z} (x+2y+az) \right] \\ & + \hat{k} \left[\frac{\partial}{\partial x} (bx-3y-z) - \frac{\partial}{\partial y} (x+2y+az) \right] \\ & = 0 \\ & \hat{i}(c+1) - \hat{j}(4-a) + \hat{k}(b-2) = 0 \end{aligned}$$

A vector field is given by $\vec{A} = (x^2 + xy^2)\hat{i} + (y^2 + x^2y)\hat{j}$. Show that the field is irrotational and find the scalar potential.

Ans. $\phi = \frac{x^3}{3} + \frac{y^3}{3} + \frac{x^2y^2}{2} + C$

Show that $\vec{A} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational. Find the velocity potential ϕ such that $\vec{A} = \nabla\phi$.

Prove that $\vec{F} = (y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is both solenoidal and irrotational

Also find the scalar potential ϕ such that $\vec{F} = \text{grad}\phi$.

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AKTU : Engineering Mathematics-1

Vector Calculus : Vector Integration

Lec-7

Today's Target

- ✓ Line Integral
- ✓ Univ. Questions
- ✓ Practice Questions



By Gulshan sir

Line Integral

Line integral of a vector function $\vec{F}(x, y, z)$ along the curve C is defined as

$$\text{Line Integral} = \int_C \vec{F} \cdot d\vec{r}$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

Note:

1. Work

If $\vec{F}$ represent the variable force acting on a particle along a curve C , then work (W) is

$$W = \int_C \vec{F} \cdot d\vec{r}$$

2. Circulation

If $\vec{F}$ represent the velocity of a fluid particle and C is closed curve

$$\text{circulation} = \oint_C \vec{F} \cdot d\vec{r}$$

3. If $\oint \vec{F} \cdot d\vec{r} = 0$, then $\vec{F}$ is said to be irrotational.

Q. 1 Find the total work done by a force $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ in moving a point from $(0, 0)$ to (a, b) along the rectangle bounded by the lines $x = 0, x = a, y = 0$ and $y = b$.

$$\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$$

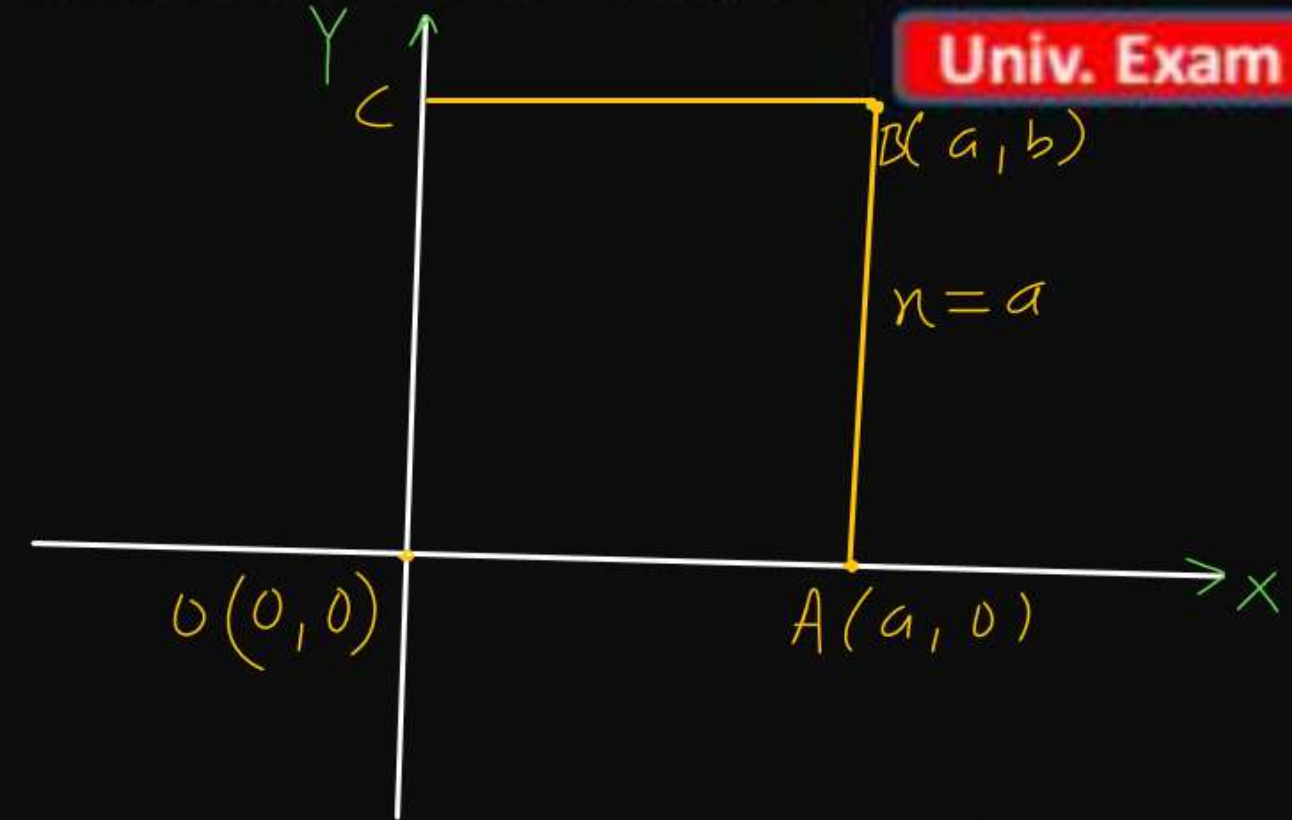
$$d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$\vec{F} \cdot d\vec{r} = [(x^2 + y^2)\hat{i} - 2xy\hat{j}] \cdot (dx\hat{i} + dy\hat{j})$$

$$\vec{F} \cdot d\vec{r} = (x^2 + y^2)dx - 2xydy \quad \text{--- (1)}$$

$$W = \int_C \vec{F} \cdot d\vec{r}$$

$$W = \int_{OA} \vec{F} \cdot d\vec{r} + \int_{AB} \vec{F} \cdot d\vec{r}$$



Along OA

$$y = 0$$

$$dy = 0$$

$$x = 0 \text{ to } x = a$$

Along AB

$$x = a$$

$$dx = 0$$

$$y = 0 \text{ to } y = b$$

Q.2 Suppose $\vec{F}(x, y, z) = x^3\hat{i} + y\hat{j} + z\hat{k}$ is the force field.

Find the work done by $\vec{F}$ along the line from $(1, 2, 3)$ to $(3, 5, 7)$.

$$\vec{F} = x^3\hat{i} + y\hat{j} + z\hat{k}$$

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\vec{F} \cdot d\vec{r} = x^3 dx + y dy + z dz \quad \text{--- ①}$$

Equation of line passing through $(1, 2, 3)$ and $(3, 5, 7)$

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

$$\frac{x - 1}{2} = \frac{y - 2}{3} = \frac{z - 3}{4} = t \text{ (say)}$$

$$x = 2t + 1 \quad dx = 2dt \quad \text{At } (1, 2, 3) \quad t = 0$$

$$y = 3t + 2 \quad dy = 3dt \quad \text{At } (3, 5, 7) \quad t = 1$$

$$z = 4t + 3 \quad dz = 4dt$$

$$W = \int_C \vec{F} \cdot d\vec{r} = \int_C x^3 dx + y dy + z dz$$

$$W = \int_0^1 (2t+1)^3 \times 2 dt + (3t+2) \times 3 dt + (4t+3) \times 4 dt$$

$$W = \int_0^1 2(2t+1)^3 dt + 3(3t+2) dt + 4(4t+3) dt$$

Q.3 If $\vec{A} = (x - y)\hat{i} + (x + y)\hat{j}$, evaluate $\oint \vec{A} \cdot d\vec{r}$ around the curve C consisting of $y = x^2$ and $x = y^2$.

$$\vec{A} = (x - y)\hat{i} + (x + y)\hat{j}$$

$$d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$\vec{A} \cdot d\vec{r} = (x - y)dx + (x + y)dy \quad \text{--- (1)}$$

Now

$$y = x^2 \quad \text{--- (2)}$$

$$x = y^2 \quad \text{--- (3)}$$

Solve (2) and (3)

$$y = (y^2)^2$$

$$y = y^4$$

$$y^4 - y = 0$$

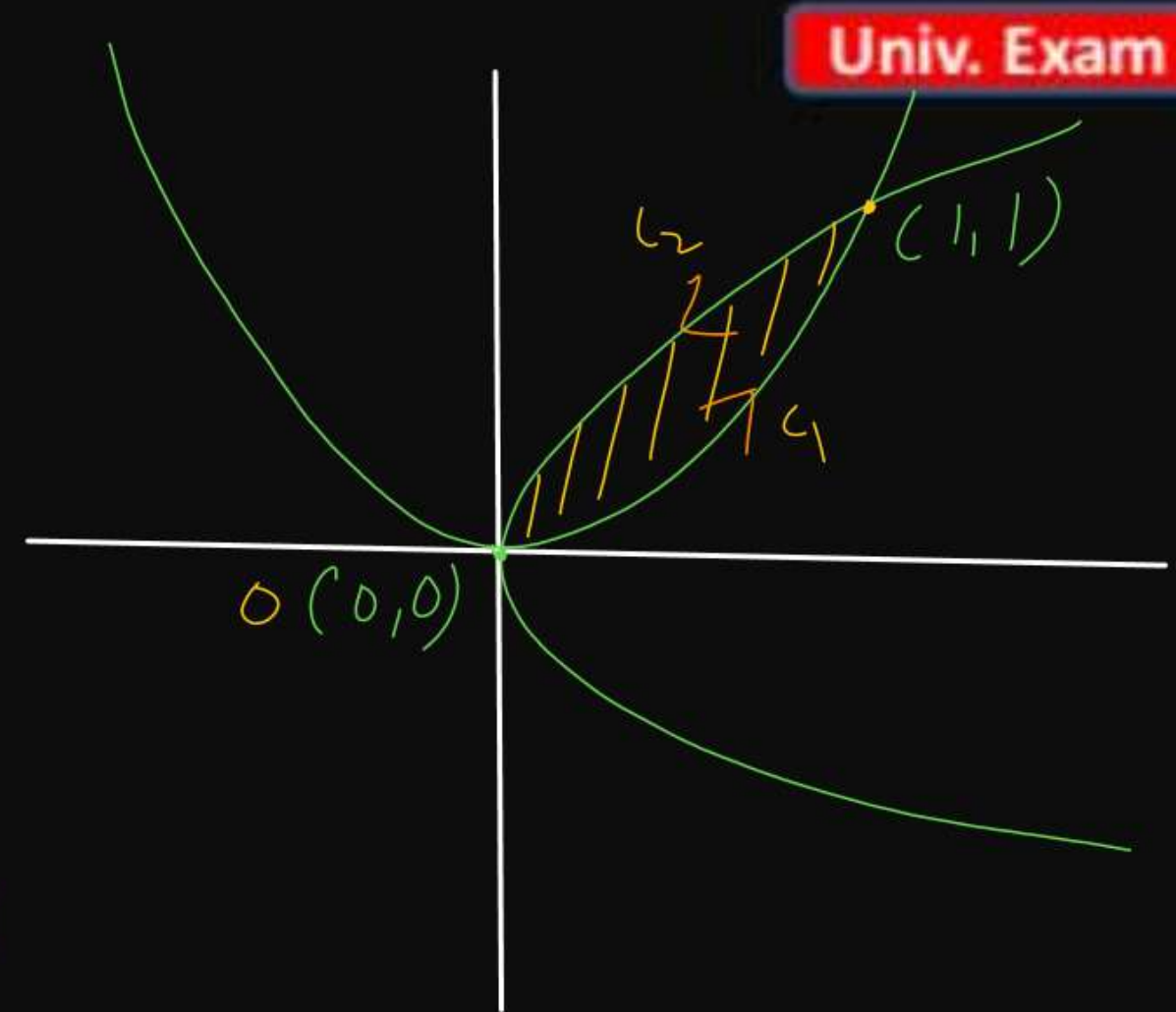
$$y(y^3 - 1) = 0$$

$$y = 0$$

$$y = 1$$

$$\text{When } y = 0, \quad x = 0$$

$$\text{When } y = 1, \quad x = 1$$



$$\oint \vec{A} \cdot d\vec{r} = \int_{C_1} \vec{A} \cdot d\vec{r} + \int_{C_2} \vec{A} \cdot d\vec{r}$$

Q.4 Find the work done in moving a particle in the force field

Univ. Exam

$$\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$$

along the curve defined by $x^2 = 4y$ and $3x^3 = 8z$ from $x = 0$ to $x = 2$.

$$\vec{F} = 3n^2\hat{i} + (2nz - y)\hat{j} + z\hat{k}$$

$$d\vec{r} = dn\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\vec{F} \cdot d\vec{r} = 3n^2 dn + (2nz - y) dy + z dz$$

Let $n = t$

$$4y = n^2$$

$$4y = t^2$$

$$y = \frac{t^2}{4}$$

$$8z = 3n^3$$

$$8z = 3t^3$$

$$z = \frac{3}{8}t^3$$

$$n = t$$

$$dn = dt$$

$$y = \frac{t^2}{4}$$

$$dy = \frac{2t}{4} = \frac{t}{2}$$

$$z = \frac{3}{8}t^3$$

$$dz = \frac{9}{8}t^2$$

When $n = 0$

$$t = 0$$

When $n = 2$

$$t = 2$$

We know that

$$W = \int_C \vec{F} \cdot d\vec{r}$$

Prac. Q. Compute the work done by the force $\vec{F} = (2y + 3)\hat{i} + xz\hat{j} + (yz - x)\hat{k}$ when it moves a particle from the point $(0, 0, 0)$ to the point $(2, 1, 1)$ along the curve $x = 2t^2, y = t, z = t^3$.

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Thank You

AKTU : Engineering Mathematics-1

Vector Calculus : Vector Integration

Lec-8

Today's Target

- ✓ **Green's Theorem**
- ✓ Univ. Questions
- ✓ Practice Questions



By Gulshan sir

If C is the regular closed curve in xy – plane and R be the region bounded by C , then

$$\int_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Where

M and N are differentiable functions inside and on C .

Q.1 Use Green's theorem to evaluate

$$\int_C (x^2 + xy)dx + (x^2 + y^2)dy$$

where C is the square formed by the lines $y = \pm 1, x = \pm 1$. ✓✓

$$\int_C (x^2 + xy)dx + (x^2 + y^2)dy$$

$$M = x^2 + xy \quad N = x^2 + y^2$$

$$\frac{\partial M}{\partial y} = x$$

$$\frac{\partial N}{\partial x} = 2x$$

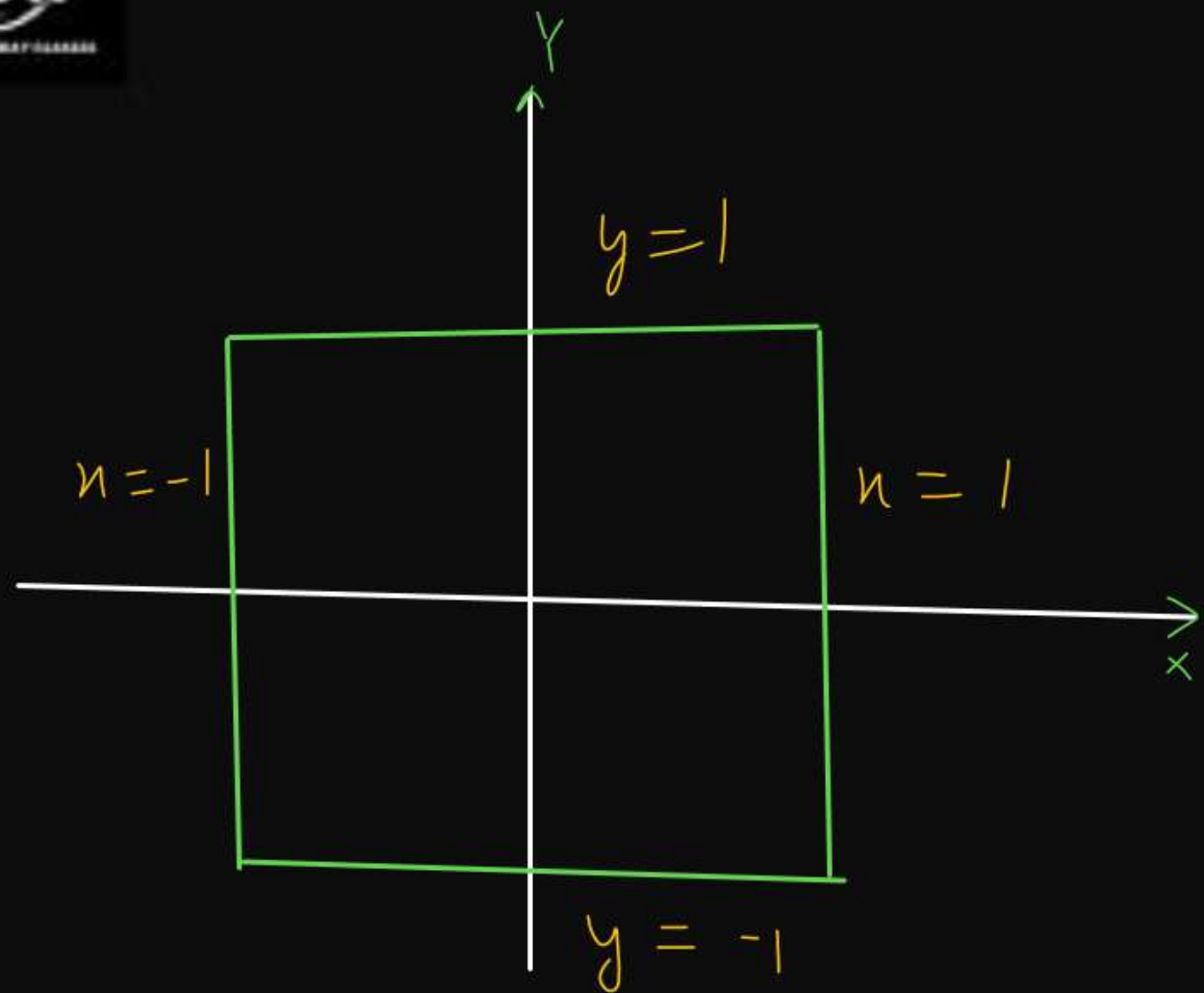
$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2x - x = x$$

By Green's theorem

$$\int_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$\int_C (x^2 + xy)dx + (x^2 + y^2)dy = \iint_R x dx dy$$

$$= \int_{x=-1}^1 \int_{y=-1}^1 x dy dx$$



$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ (even)}$$

$$= 0 \text{ (odd)}$$

$$= \int_{x=-1}^1 x dx \int_{y=-1}^1 dy = \int_{-1}^1 x dx (y) \Big|_{-1}^1$$

$$= \int_{-1}^1 x dx (1+1) = 2 \int_{-1}^1 x dx$$

$$= 2 \int_{-1}^1 x dx$$

$$= 2 \times 0$$

$$= 0$$

Q.2 Using Green's theorem to evaluate $\int_C x^2 y dx + x^2 dy$ where

Univ. Exam

C is the boundary described anti-clockwise of the triangle with vertices $(0, 0)$; $(1, 0)$; $(1, 1)$.

$$\int_C x^2 y dx + x^2 dy$$

$$M = x^2 y \quad | \quad N = x^2$$

$$\frac{\partial M}{\partial y} = x^2 \quad | \quad \frac{\partial N}{\partial x} = 2x$$

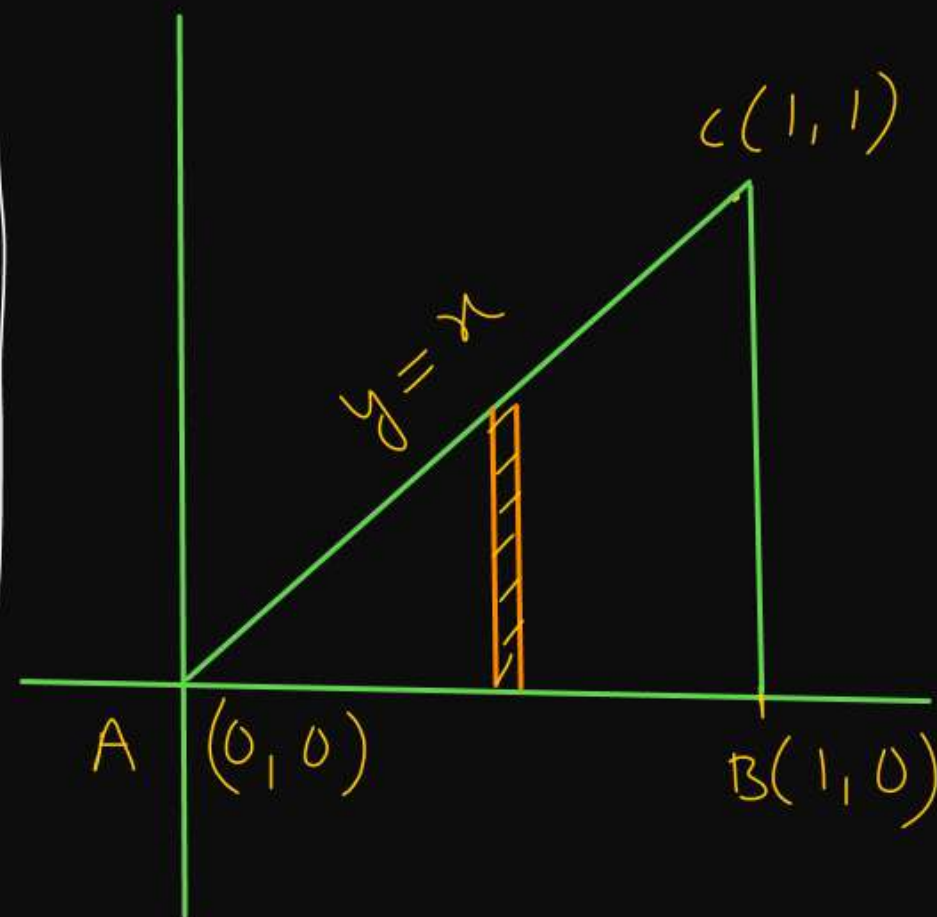
$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2x - x^2$$

By Green's theorem

$$\int_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$\int_C x^2 y dx + x^2 dy = \iint_R (2x - x^2) dx dy$$

$$= \int_{x=0}^1 \int_{y=0}^x (2x - x^2) dx dy$$



$$y = mx$$

$$y = x$$

$$= \int_0^1 \int_0^u (2u - u^2) du dy$$

$$= \int_0^1 (2u - u^2) du \int_0^u dy$$

$$= \int_0^1 (2u - u^2) (y)_0^u du$$

$$= \int_0^1 (2u - u^2) u du$$

$$= \int_0^1 (2u^2 - u^3) du$$

$$= \left(\frac{2u^3}{3} - \frac{u^4}{4} \right)_0^1$$

$$= \frac{2}{3} - \frac{1}{4}$$

$$= \frac{8 - 3}{12}$$

$$= \frac{5}{12}$$

Q.3 Apply Green's theorem to evaluate $\int_C [(2x^2 - y^2)dx + (x^2 + y^2)dy]$

Univ. Exam

where C is the boundary of the area enclosed by the x -axis and the upper half of the circle $x^2 + y^2 = a^2$.

$$\int_C (2x^2 - y^2)dx + (x^2 + y^2)dy$$

$$M = 2x^2 - y^2$$

$$N = x^2 + y^2$$

$$\frac{\partial M}{\partial y} = -2y$$

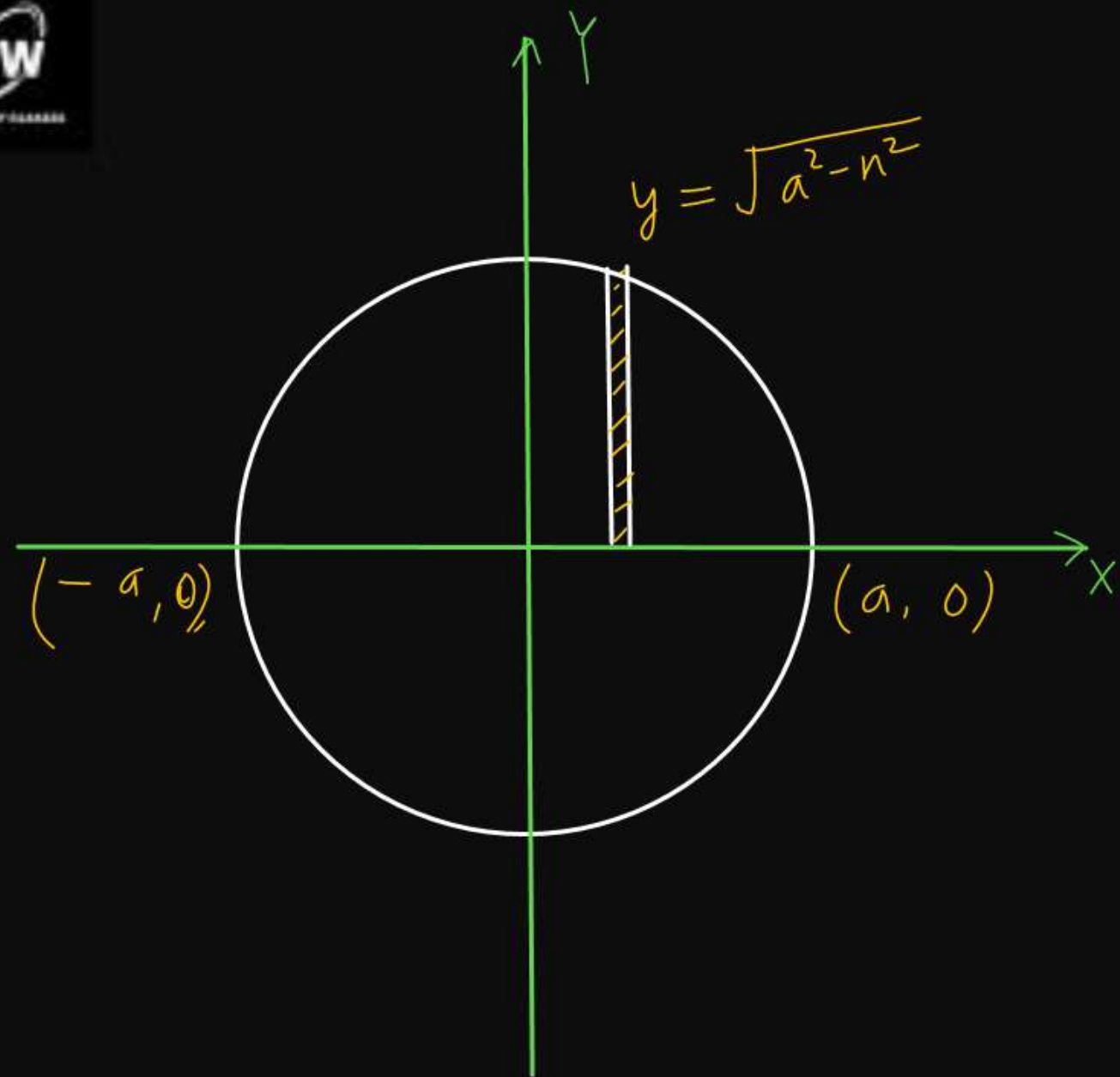
$$\frac{\partial N}{\partial x} = 2x$$

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2x + 2y$$

By Green's theorem

$$\int_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$\begin{aligned} \int_C (2x^2 - y^2)dx + (x^2 + y^2)dy &= \iint_R 2(x + y) dx dy \\ &= \int_{x=-a}^a \int_{y=0}^{\sqrt{a^2-x^2}} 2(x + y) dx dy \end{aligned}$$



$$x^2 + y^2 = a^2$$

$$y = \sqrt{a^2 - x^2}$$

$$= \int_{-a}^a \int_0^{\sqrt{a^2 - x^2}} 2(x + y) dx dy$$

$$= 2 \int_{-a}^a \left(xy + \frac{y^2}{2} \right) \Big|_0^{\sqrt{a^2 - x^2}} dx$$

$$= 2 \int_{-a}^a \left(x \sqrt{a^2 - x^2} + \frac{a^2 - x^2}{2} \right) dx$$

$$= 2 \int_{-a}^a x \sqrt{a^2 - x^2} dx + \int_{-a}^a (a^2 - x^2) dx$$

$$= 0 + 2 \int_0^a (a^2 - x^2) dx$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \quad (\text{even})$$

$$= 0 \quad (\text{odd})$$

$$= 2 \left(a^2 x - \frac{x^3}{3} \right) \Big|_0^a$$

$$= 2 \left(a^3 - \frac{a^3}{3} \right)$$

$$= 2 \times \frac{2a^3}{3}$$

$$= \frac{4a^3}{3}$$

Q.4 Using Green's theorem to evaluate $\int_C 2y^2 dx + 3x dy$ where

Univ. Exam

C is the boundary of the closed region bounded by $y = x$, and $y = x^2$.

$$\int_C 2y^2 dx + 3x dy$$

$$M = 2y^2$$

$$\frac{\partial M}{\partial y} = 4y$$

$$N = 3x$$

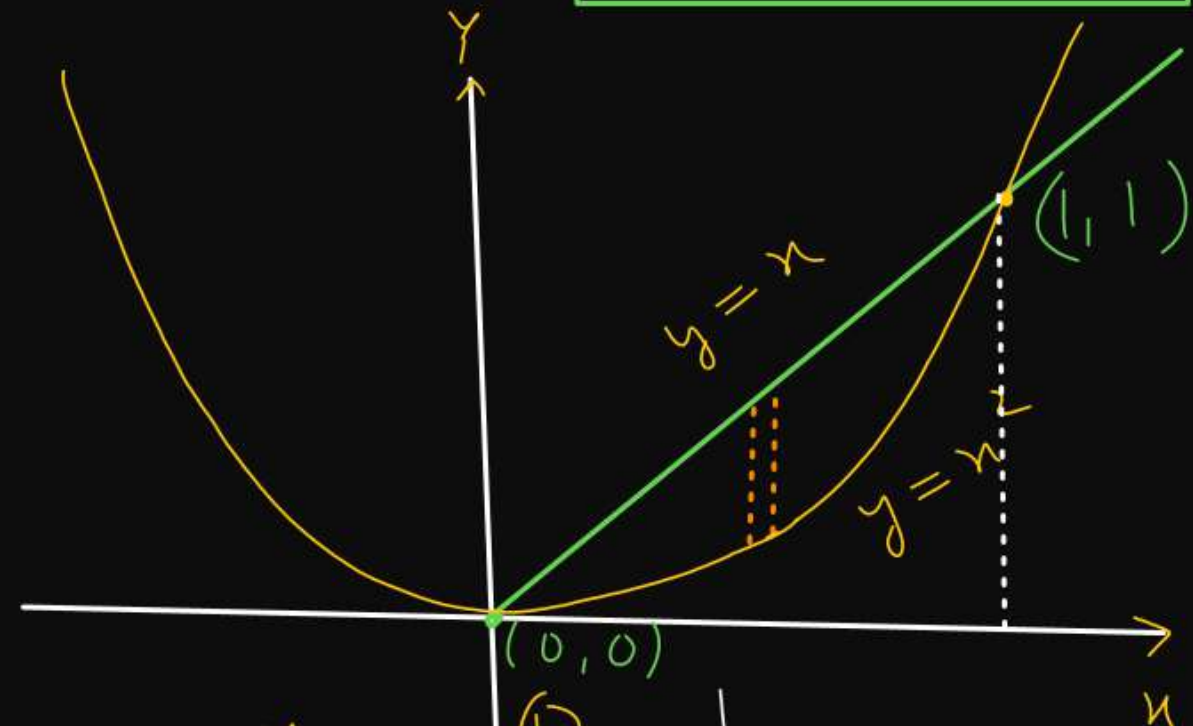
$$\frac{\partial N}{\partial x} = 3$$

$$\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y} = 3 - 4y$$

By Green's theorem

$$\int_C M dx + N dy = \iint_R \left(\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y} \right) dx dy$$

$$\begin{aligned} \int_C 2y^2 dx + 3x dy &= \iint_R (3 - 4y) dx dy \\ &= \int_{x=0}^1 \int_{y=x^2}^x (3 - 4y) dy dx \end{aligned}$$



$$\begin{aligned} y &= x & \text{--- (1)} \\ y &= x^2 & \text{--- (2)} \\ \text{solve (1) and (2)} \\ x^2 &= x \\ x^2 - x &= 0 \end{aligned}$$

$$\begin{aligned} x(x-1) &= 0 \\ x &= 0 \\ x &= 1 \\ y &= 0 \\ y &= 1 \end{aligned}$$

$$= \int_0^1 \int_{n^2}^n (3 - 4y) dy dn$$

$$= \int_0^1 \left(3y - 2y^2 \right)_{n^2}^n dn$$

$$= \int_0^1 (3n - 2n^2 - 3n^2 + 2n^4) dn$$

$$= \int_0^1 (3n - 5n^2 + 2n^4) dn$$

$$= \left(\frac{3n^2}{2} - \frac{5n^3}{3} + \frac{2n^5}{5} \right)_0^1$$

$$= \frac{3}{2} - \frac{5}{3} + \frac{2}{5}$$

$$= \frac{45 - 50 + 12}{30}$$

$$= \frac{7}{30}$$

Q.5 Verify Green's theorem in the plane for $\int_C (3x^2 - 8y^2)dx + (4x - 6xy)dy$ where C is the region bounded by the parabolas $y = \sqrt{x}$, and $y = x^2$.

$$\int_C (3x^2 - 8y^2)dx + (4x - 6xy)dy$$

$$M = 3x^2 - 8y^2$$

$$\frac{\partial M}{\partial y} = -16y$$

$$N = 4x - 6xy$$

$$\frac{\partial N}{\partial x} = 4 - 6y$$

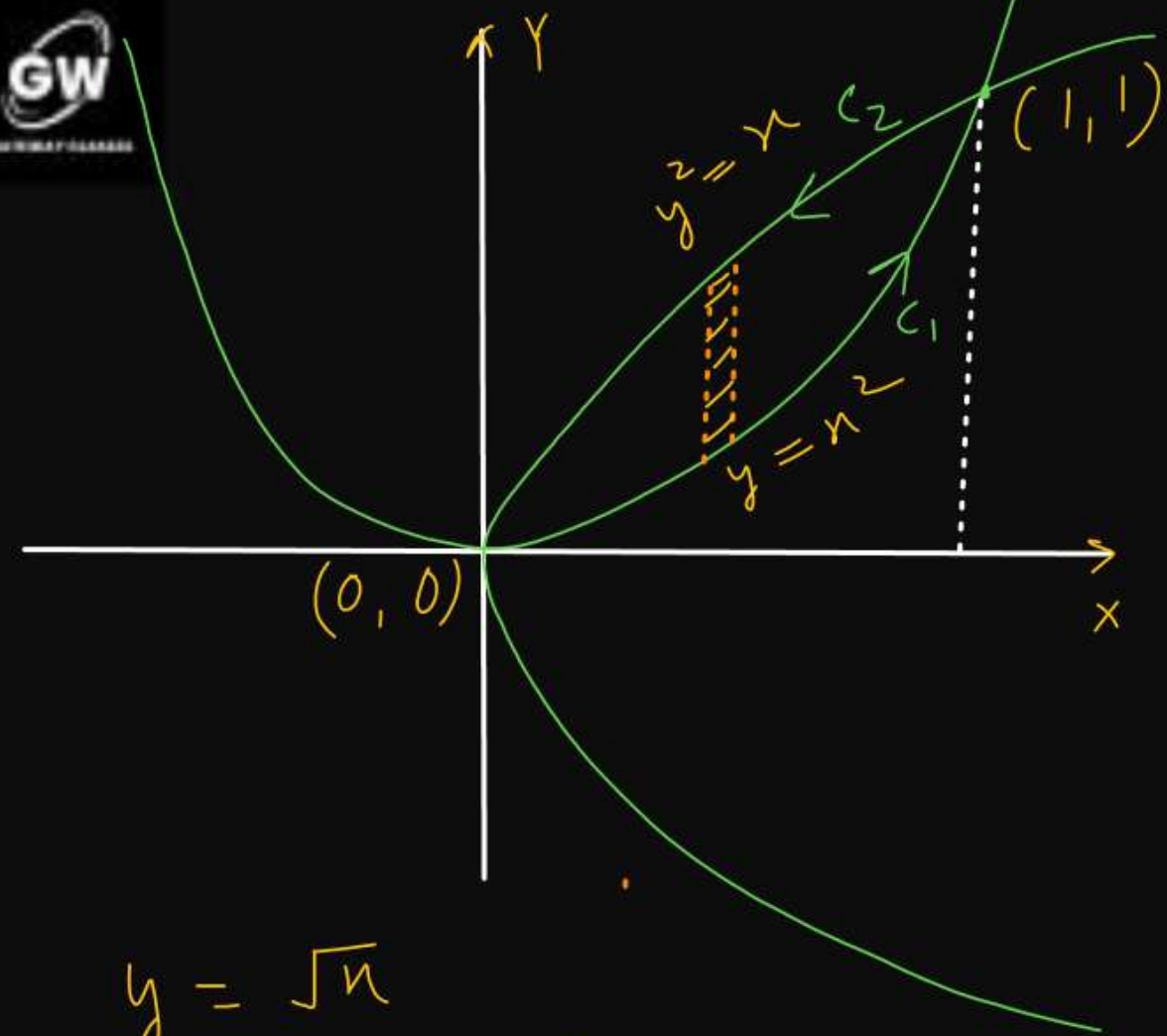
$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 4 - 6y - (-16y)$$

$$= 10y + 4$$

By Green's theorem

$$\int_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

RHS



$$y = \sqrt{x}$$

$$y^2 = x \quad \text{--- (1)}$$

$$x^2 = y \quad \text{--- (2)}$$

$$= \int \int_R (10y + 4) dx dy$$

$$= \int_{x=0}^1 \int_{y=x^2}^{\sqrt{x}} (10y + 4) dx dy$$

$$= \int_0^1 (5y^2 + 4y) \Big|_{x^2}^{\sqrt{x}}$$

$$= \int_0^1 (5x + 4\sqrt{x} - 5x^4 - 4x^2) dx$$

$$= \int_0^1 (5x + 4x^{1/2} - 5x^4 - 4x^2) dx$$

$$= \left(\frac{5x^2}{2} + \frac{8}{3} x^{3/2} - x^5 - \frac{4x^3}{3} \right) \Big|_0^1$$

$$= \frac{5}{2} + \frac{8}{3} - 1 - \frac{4}{3}$$

$$= \frac{15 + 16 - 6 - 8}{6}$$

$$= \frac{17}{6}$$

LMS

$$\int M dx + N dy = \int_C (3x^2 - 8y^2) dx + (4x - 6xy) dy$$

$$= \int_{C_1 + C_2} (3x^2 - 8y^2) dx + (4x - 6xy) dy$$

$$= \int_{C_1} (3x^2 - 8y^2) dx + (4x - 6xy) dy + \int_{C_2} (3x^2 - 8y^2) dx + (4x - 6xy) dy$$

For C_1
 $y = x^2$
 $dy = 2x dx$
 $x = 0 \text{ to } x = 1$

For C_2
 $y^2 = x$
 $2y dy = dx$
 $y = 0 \text{ to } y = 1$

$$= \int_0^1 (3x^2 - 8x^4) dx + (4x - 6x^3) 2x dx +$$

$$\int_1^0 (3y^4 - 8y^2) 2y dy + (4y^2 - 6y^3) dy$$

Prac. Q. 1 Evaluate by Green's theorem $\int (Cosx siny - xy)dx + sinx cosy dy$

where C is the circle $x^2 + y^2 = 1$

Ans. 0

Prac. Q. 2 Verify Green's theorem in the plane for $\int (xy + y^2)dx + x^2 dy$

where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.

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AKTU : Engineering Mathematics-1

Vector Calculus : Vector Integration

Lec-9

Today's Target

- ✓ **Stoke's Theorem**
- ✓ Univ. Questions
- ✓ Practice Questions

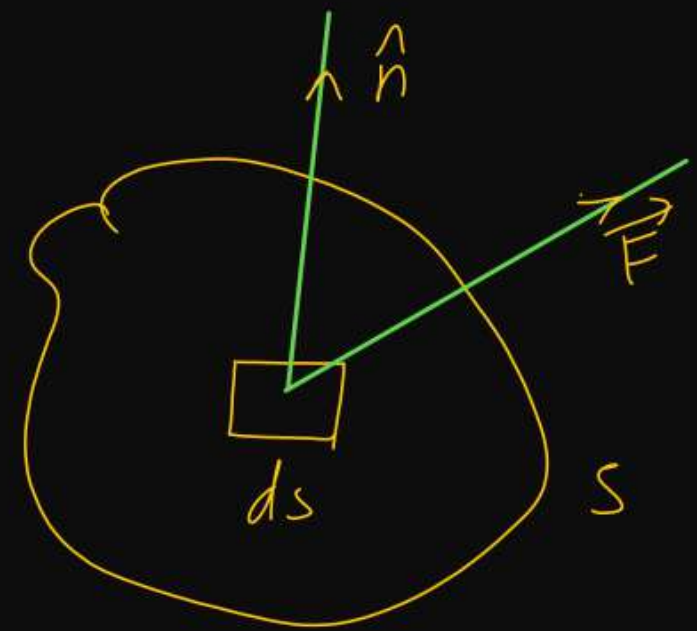


By Gulshan sir

Surface Integral

Surface Integral of $\vec{F}$ over surface S is given as

$$\text{Surface integral} = \iint_R \vec{F} \cdot \hat{n} \, ds$$



Where

$$(i) \quad \hat{n} = \frac{\nabla \phi}{|\nabla \phi|}$$

(ii) If R be the projection of S on xy – plane then

$$ds = \frac{dx \, dy}{|\hat{n} \cdot \hat{k}|}$$

(iii) If R be the projection of S on yz – plane then

$$ds = \frac{dz \, dy}{|\hat{n} \cdot \hat{j}|}$$

(iv) If R be the projection of S on zx – plane then

$$ds = \frac{dx \, dz}{|\hat{n} \cdot \hat{i}|}$$

Stoke's Theorem (Relation between line and surface integral)

If S is an open surface bounded by a closed curve C and $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ is any vector point function having continuous first order partial derivatives then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

Where $\hat{n}$ is a unit normal vector at any point of S

Q.1 Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by Stoke's theorem where $\vec{F} = y^2\hat{i} + x^2\hat{j} - (x+z)\hat{k}$ and C is the boundary of the triangle with vertices at $(0,0,0)$, $(1,0,0)$ and $(1,1,0)$.

$$\vec{F} = y^2\hat{i} + x^2\hat{j} - (x+z)\hat{k}$$

$$\text{curl } \vec{F} = \nabla \times \vec{F}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x^2 & -x-z \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y}(-x-z) - \frac{\partial}{\partial z}(x^2) \right] - \hat{j} \left[\frac{\partial}{\partial x}(-x-z) - \frac{\partial}{\partial z}(y^2) \right] + \hat{k} \left[\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(y^2) \right]$$

$$= \hat{i}(0-0) - \hat{j}(-1-0) + \hat{k}(2x-2y)$$

Univ. Exam

$$= 0\hat{i} + \hat{j} + 2(x-y)\hat{k}$$

$$\text{curl } \vec{F} = 0\hat{i} + \hat{j} + 2(x-y)\hat{k}$$

Since z coordinates are zero

$\therefore$ Triangle formed in $x-y$ plane

$$\hat{n} = \hat{k}$$

$$ds = \frac{dn dy}{|\hat{n} \cdot \hat{k}|}$$

$$= \frac{dn dy}{|\hat{k} \cdot \hat{k}|}$$

$$= \frac{dn dy}{|\hat{k}|^2}$$

$$ds = dn dy$$

By stoke's theorem

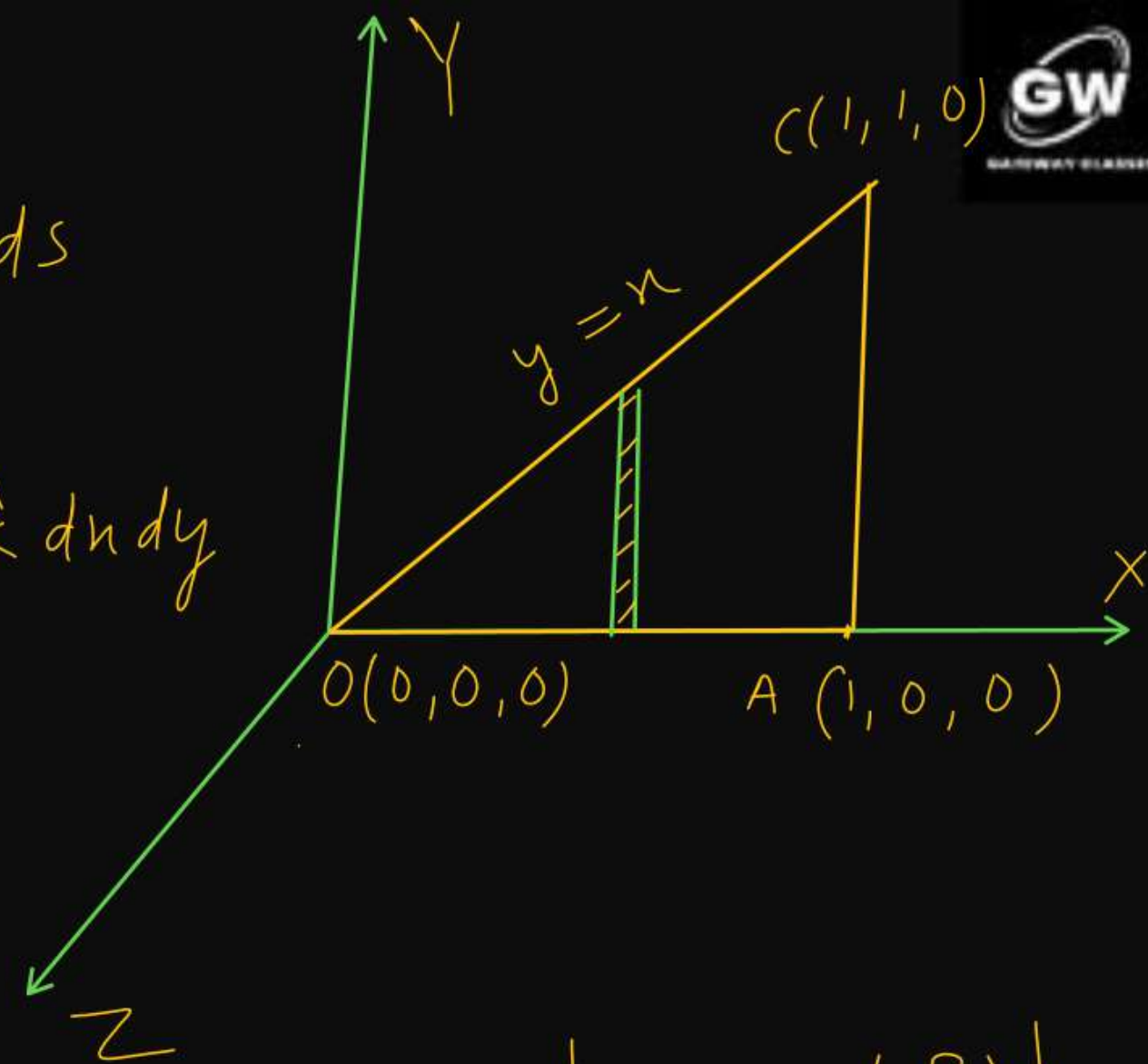
$$\oint \vec{F} \cdot d\vec{r} = \iint_S (\text{curl } \vec{F}) \cdot \hat{n} ds$$

$$= \iint \{ \hat{j} + 2(n-y)\hat{k} \} \cdot \hat{k} dn dy$$

$$= \int_{n=0}^1 \int_{y=0}^n 2(n-y) dn dy$$

$$= 2 \int_{n=0}^1 \left(ny - \frac{y^2}{2} \right)_0^n dn = 2 \int_{n=0}^1 \left(n^2 - \frac{n^2}{2} \right) dn = 2 \int_0^1 \frac{n^2}{2} dn = \left(\frac{n^3}{3} \right)_0^1$$

$$\oint \vec{F} \cdot d\vec{r} = \frac{1}{3}$$



Q.2 Verify Stoke's theorem for $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ taken round the

Univ. Exam

rectangle bounded by the lines $x = \pm a$, $y = 0$, $y = b$.

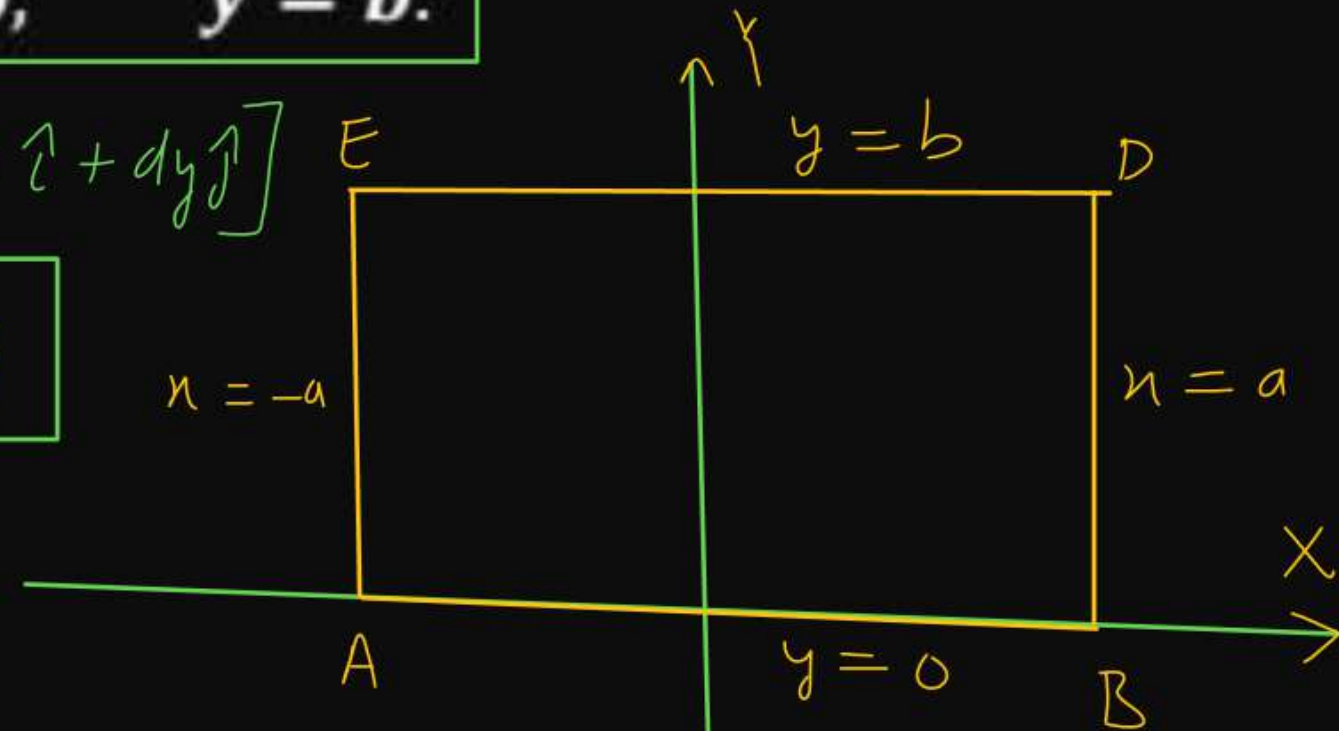
$$\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$$

$$\vec{F} \cdot d\vec{r} = [(x^2 + y^2)\hat{i} - 2xy\hat{j}] \cdot [dx\hat{i} + dy\hat{j}]$$

Stoke's theorem

$$\vec{F} \cdot d\vec{r} = (x^2 + y^2)dx - 2xydy$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\text{curl } \vec{F}) \cdot \hat{n} ds$$



LHS

$$\oint_C \vec{F} \cdot d\vec{r} = \oint_{AB} \vec{F} \cdot d\vec{r} + \oint_{BD} \vec{F} \cdot d\vec{r} + \oint_{DE} \vec{F} \cdot d\vec{r} + \oint_{EA} \vec{F} \cdot d\vec{r}$$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{-a}^a x^2 dx + \int_0^b (-2ay) dy + \int_a^{-a} (x^2 + b^2) dx + \int_b^0 2ay dy$$

$$\begin{array}{l} \text{AB} \\ y = 0 \\ dy = 0 \end{array}$$

$$\begin{array}{l} x = -a \\ \text{to} \\ x = a \end{array}$$

$$\begin{array}{l} \text{BD} \\ x = a \\ dx = 0 \end{array}$$

$$\begin{array}{l} y = 0 \\ \text{to} \\ y = b \end{array}$$

$$\begin{array}{l} \text{DE} \\ y = b \\ dy = 0 \end{array}$$

$$\begin{array}{l} x = a \\ \text{to} \\ x = -a \end{array}$$

$$\text{EA}$$

$$\begin{array}{l} x = -a \\ dx = 0 \\ y = b \\ \text{to} \\ y = 0 \end{array}$$

$$\oint \vec{F} \cdot d\vec{r} = \left(\frac{u^3}{3} \right)_a^a - 2a \left(\frac{y^2}{2} \right)_0^b + \left(\frac{n^3}{3} + b^2 n \right)_a^a + 2a \left(\frac{y^2}{2} \right)_b^0$$

$$= \cancel{\frac{a^3}{3}} + \cancel{\frac{a^3}{3}} - ab^2 - \cancel{\frac{a^3}{3}} - ab^2 - \cancel{\frac{a^3}{3}} - ab^2 - ab^2$$

$$= -4ab^2$$

RHS

$$\iint_S \text{curl } \vec{F} \cdot \hat{n} ds$$

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2+y^2 & -2xy & 0 \end{vmatrix}$$

$$\begin{aligned} \text{curl } \vec{F} &= \hat{i}(0+0) - \hat{j}(0-0) + \hat{k}(-2y-2y) \\ &= 0\hat{i} + 0\hat{j} - 4y\hat{k} = -4y\hat{k} \end{aligned}$$

$$\hat{n} = \hat{k}$$

$$\text{curl } \vec{F} \cdot \hat{n} = (-4y\hat{k}) \cdot \hat{k} = -4y$$

$$ds = \frac{dx dy}{|\hat{n} \cdot \hat{k}|}$$

$$ds = dx dy$$

$$\int \int_S \text{curl } \vec{F} \cdot \hat{n} \, ds = \int_{n=-a}^a \int_{y=0}^b (-4y) \, dn \, dy$$

$$= -4 \int_{-a}^a \left(\frac{y^2}{2} \right)_0^b \, dn$$

$$= -4 \int_{-a}^a \frac{b^2}{2} \, dn$$

$$= -2b^2 \int_{-a}^a \, dn$$

$$= -2b^2 (n)_{-a}^a = -2b^2 (a + a)$$

$$\int \int_S \text{curl } \vec{F} \cdot \hat{n} \, ds = -4ab^2$$

$$\text{LHS} = \text{RHS}$$

Hence Stoke's theorem
is verified

Q.3 Verify Stoke's theorem for $\vec{F} = (y - z + 2)\hat{i} + (yz + 4)\hat{j} - xz\hat{k}$ over the surface of cube $x = 0, y = 0, z = 0, x = 2, y = 2, z = 2$ above the XOY -plane (open the bottom).

$$\vec{F} = (y - z + 2)\hat{i} + (yz + 4)\hat{j} - xz\hat{k}$$

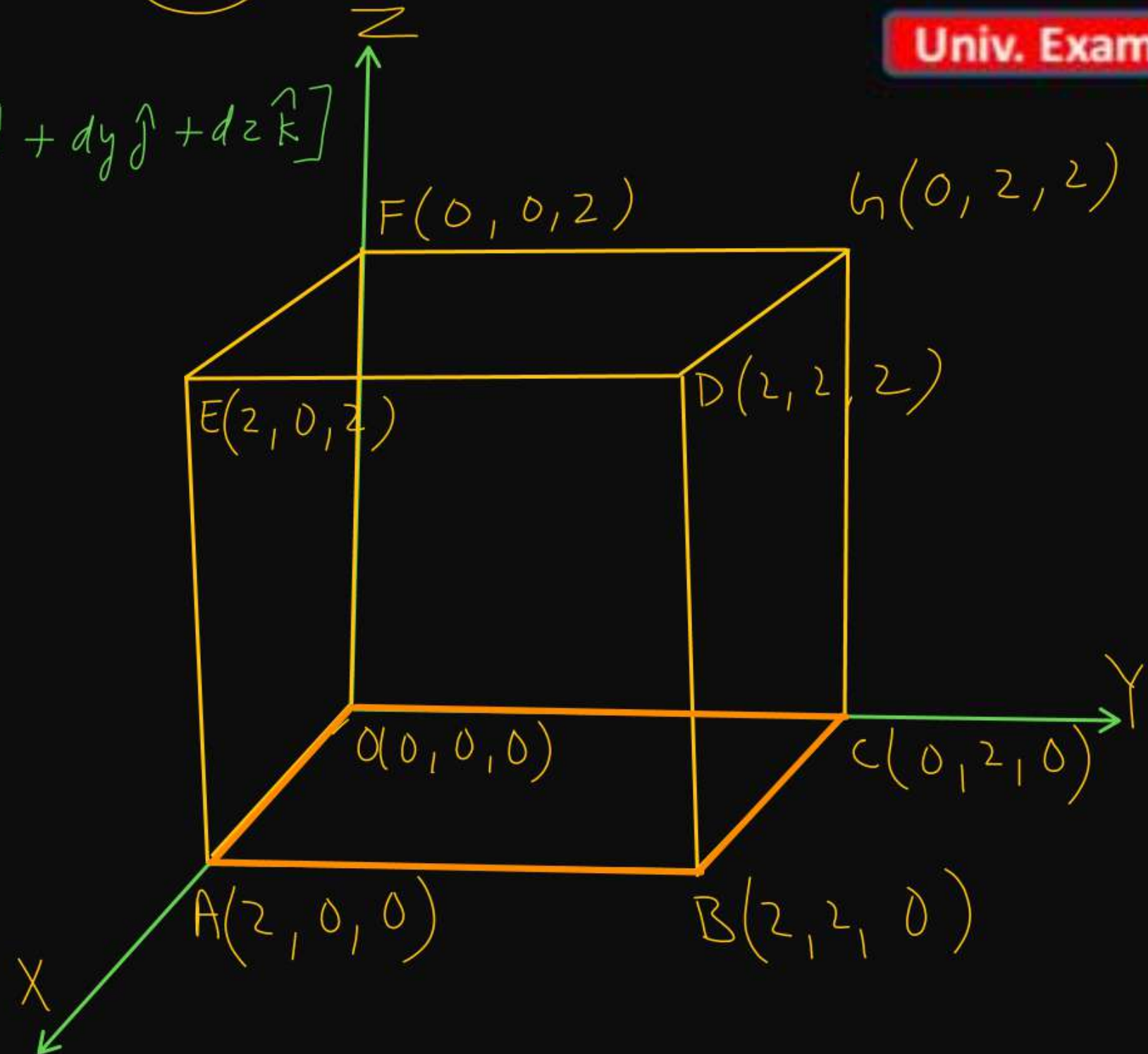
$$\vec{F} \cdot d\vec{r} = [(y - z + 2)\hat{i} + (yz + 4)\hat{j} - xz\hat{k}] \cdot [dx\hat{i} + dy\hat{j} + dz\hat{k}]$$

$$\vec{F} \cdot d\vec{r} = (y - z + 2)dx + (yz + 4)dy - xzdz$$

Stoke's theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iiint_S \text{curl } \vec{F} \cdot \hat{n} ds$$

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LMS

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{OA} \vec{F} \cdot d\vec{r} + \int_{AB} \vec{F} \cdot d\vec{r} + \int_{BC} \vec{F} \cdot d\vec{r} + \int_{CO} \vec{F} \cdot d\vec{r}$$

For OA

$$y = 0$$

$$z = 0$$

$$dy = 0$$

$$dz = 0$$

$$\begin{cases} x = 0 \\ \text{to} \\ x = 2 \end{cases}$$

For AB

$$z = 0$$

$$x = 2$$

$$dz = 0$$

$$dx = 0$$

$$y = 0$$

to

$$y = 2$$

For BC

$$z = 0$$

$$y = 2$$

$$dz = 0$$

$$dy = 0$$

$$x = 2$$

to

$$x = 0$$

For CO

$$x = 0$$

$$z = 0$$

$$dx = 0$$

$$dz = 0$$

$$y = 2$$

to

$$y = 0$$

$$\vec{F} \cdot d\vec{r} = (y - z + 2)dx + (yz + 4)dy - xz dz$$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^2 2 dx + \int_0^2 4 dy + \int_2^0 4 dx + \int_2^0 4 dy$$

$$= 2(x)_0^2 + 4(y)_0^2 + 4(x)_2^0 + 4(y)_2^0$$

$$= 2(2 - 0) + 4(2 - 0) + 4(0 - 2) + 4(0 - 2)$$

$$= 4 + \cancel{8} - \cancel{8} - 8$$

$$\oint_C \vec{F} \cdot d\vec{r} = -4$$

RHS

$$\iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

$$\text{curl } \vec{F} = \nabla \times \vec{F}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (y-z+2) & (yz+4) & -xz \end{vmatrix}$$

$$= \hat{i}(0-y) - \hat{j}(-z+1) + \hat{k}(0-1)$$

$$\boxed{\text{curl } \vec{F} = -y\hat{i} + (z-1)\hat{j} - \hat{k}}$$

For ABDE

$$\hat{n} = \hat{i}, \quad n = 2$$

$$ds = \frac{dy \, dz}{|\hat{n} \cdot \hat{i}|} = \frac{dy \, dz}{|\hat{i} \cdot \hat{i}|} = dy \, dz$$

$$\int_{z=0}^2 \int_{y=0}^2 [-y\hat{i} + (z-1)\hat{j} - \hat{k}] \cdot (\hat{i}) \, dy \, dz$$

$$= \int_0^2 \int_0^2 (-y) \, dy \, dz$$

$$= - \int_0^2 \left(\frac{y^2}{2} \right)_0^2 \, dz$$

$$= - \int_0^2 2 \, dz = -2(z)_0^2$$

$$= -2(2-0)$$

$$= -4$$

For COF6

$$\hat{n} = -\hat{i}$$

$$ds = \frac{dy dz}{|\hat{n} \cdot \hat{i}|}$$

$$= \frac{dy dz}{|-\hat{i} \cdot \hat{i}|}$$

$$= dy dz$$

$$x = 0$$

$$\iint_S \text{curl } \vec{F} \cdot \hat{n} ds = \int \int [y\hat{i} + (z-1)\hat{j} - \hat{k}] \cdot (-\hat{i}) dy dz$$

$$= \int_{z=0}^2 \int_{y=0}^2 y dy dz = \int_0^2 \left(\frac{y^2}{2} \right)_0^2 dz = \int_0^2 2 dz$$

$$= 2(z)_0^2 = 2(2-0) = 4$$

Q.4 Apply Stoke's theorem to evaluate

$$\int_C [(x+y)dx + (2x-z)dy + (y+z)dz]$$

where C is the boundary of the triangle with vertices at $(2, 0, 0)$, $(0, 3, 0)$ and $(0, 0, 6)$.

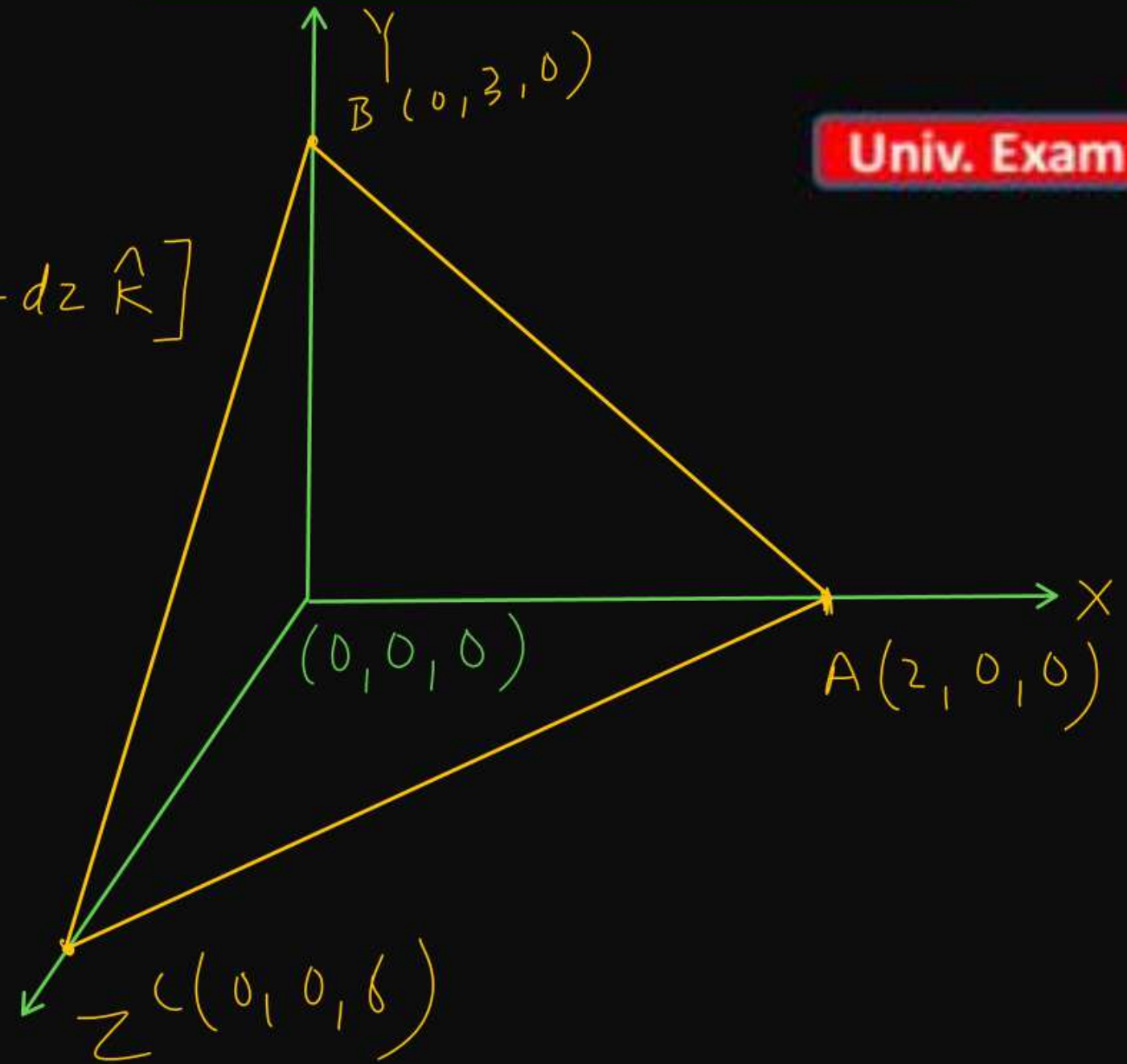
$$\int_C (x+y)dx + (2x-z)dy + (y+z)dz$$

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$$= \int_C [(x+y)\hat{i} + (2x-z)\hat{j} + (y+z)\hat{k}] [dx\hat{i} + dy\hat{j} + dz\hat{k}]$$

$$= \int_C \vec{F} \cdot d\vec{r}$$

$$\vec{F} = (x+y)\hat{i} + (2x-z)\hat{j} + (y+z)\hat{k}$$



Stoke's theorem

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

Equation of surface (Plane) ABC

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\frac{x}{2} + \frac{y}{3} + \frac{z}{6} = 1$$

$$\text{Let } \phi = \frac{x}{2} + \frac{y}{3} + \frac{z}{6} - 1$$

Ans 21

P.Q. 1 Verify Stoke's theorem for $\vec{F} = x^2\hat{i} + xy\hat{j}$ integrated round the square whose sides are $x = 0, y = 0, x = a, y = a$ in the plane $z = 0$.

P.Q.2 Verify Stoke's theorem for $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$ and surfaces S is the portion of the sphere $x^2 + y^2 + z^2 = 1$ above the xy - plane .

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P.Q.3 Verify Stoke's theorem for the vector field $\vec{F} = (x^2 - y^2)\hat{i} + 2xy\hat{j}$ integrated round the rectangle in the plane $z = 0$ and bounded by the lines $x = 0, y = 0, x = a, y = b$.

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P.Q.4 Verify Stoke's theorem for the vector field $\vec{F} = (2y + z, x - z, y - x)$ taken over the triangle ABC cut from the plane $x + y + z = 1$ by the coordinates.

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Thank You

AKTU : Engineering Mathematics-1

Vector Calculus : Vector Integration

Lec-10

Today's Target

- ✓ Gauss Divergence Theorem
- ✓ Univ. Questions
- ✓ Practice Questions



By Gulshan sir

If $\vec{F}$ is a vector point function having continuous first order partial derivatives in the region V bounded by a closed surface S then

$$\int \int_S \vec{F} \cdot \hat{n} \, ds = \int \int \int_V \text{div} \vec{F} \, dv$$

surface integral

volume integral

Where $\hat{n}$ is the outward drawn unit normal vector to the surface S

Q.1 Find $\iint_S \vec{F} \cdot \hat{n} ds$ where $\vec{F} = (2x + 3z)\hat{i} - (xz + y)\hat{j} + (y^2 + 2z)\hat{k}$

and S is the surface of sphere having centre at $(3, -1, 2)$ and radius 3.

$$\vec{F} = (2x + 3z)\hat{i} - (xz + y)\hat{j} + (y^2 + 2z)\hat{k}$$

$$\text{div } \vec{F} = \nabla \cdot \vec{F}$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (2x + 3z)\hat{i} - (xz + y)\hat{j} + (y^2 + 2z)\hat{k}$$

$$= \frac{\partial}{\partial x}(2x + 3z) - \frac{\partial}{\partial y}(xz + y) + \frac{\partial}{\partial z}(y^2 + 2z)$$

$$\text{div } \vec{F} = 2 - 1 + 2$$

$$\boxed{\text{div } \vec{F} = 3}$$

By Gauss Divergence Theorem

$$\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \text{div } \vec{F} dv$$

$$= \iiint_V 3 dv = 3 \iiint_V dv$$

$$= 3V$$

Here V is the volume of sphere

$$V = \frac{4}{3} \pi r^3$$

Q.2 Evaluate $\iint_S \vec{r} \cdot \hat{n} ds$ where S is a closed surface and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\text{div } \vec{r} = \nabla \cdot \vec{r}$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x\hat{i} + y\hat{j} + z\hat{k})$$

$$= 1 + 1 + 1$$

$$\boxed{\text{div } \vec{r} = 3}$$

By Gauss Divergen Theorem

$$\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \text{div } \vec{F} dv$$

$$\text{Put } \vec{F} = \vec{r}$$

$$\iint_S \vec{r} \cdot \hat{n} ds = \iiint_V \text{div } \vec{r} dv$$

$$= \iiint_V 3 dv = 3 \iiint_V dv$$

$$\boxed{\iint_S \vec{r} \cdot \hat{n} ds = 3V}$$

Q.3 Evaluate $\iint_S (y^2 z^2 \hat{i} + z^2 x^2 \hat{j} + z^2 y^2 \hat{k}) \cdot \hat{n} ds$, where S is the part of the sphere $x^2 + y^2 + z^2 = 1$ above the xy plane and bounded by this plane.

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$$\iint_S (y^2 z^2 \hat{i} + z^2 x^2 \hat{j} + z^2 y^2 \hat{k}) \cdot \hat{n} ds$$

Where

$$\vec{F} = y^2 z^2 \hat{i} + z^2 x^2 \hat{j} + z^2 y^2 \hat{k}$$

$$\text{div } \vec{F} = \nabla \cdot \vec{F}$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (y^2 z^2 \hat{i} + z^2 x^2 \hat{j} + z^2 y^2 \hat{k})$$

$$= \frac{\partial}{\partial x}(y^2 z^2) + \frac{\partial}{\partial y}(z^2 x^2) + \frac{\partial}{\partial z}(z^2 y^2)$$

$$\text{div } \vec{F} = 0 + 0 + 2zy^2$$

$$\text{div } \vec{F} = 2zy^2$$

By Gauss Divergence theorem

$$\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \text{div } \vec{F} dv$$

$$= \iiint_V 2zy^2 dv$$

change into spherical coordinates

P.Q. 1 Evaluate $\iint_S (yz\hat{i} + zx\hat{j} + xy\hat{k}) \cdot d\vec{S}$ where S is the surface of the sphere

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$x^2 + y^2 + z^2 = a^2$ in the first octant.

Ans. 0

P.Q.2 Verify divergence theorem for $\vec{F} = x^3\hat{i} - y^3\hat{j} + z^3\hat{k}$ taken over the surface of sphere $x^2 + y^2 + z^2 = a^2$.

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Q.4 Use divergence theorem to evaluate the surface integral

$$\iint_S (x dy dz + y dz dx + z dx dy) \text{ where } s \text{ is the portion of the plane } x + 2y + 3z = 6$$

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$$\iint_S x dy dz + y dz dx + z dx dy$$

$$= \iint_S (x \hat{i} + y \hat{j} + z \hat{k}) \cdot (dy dz \hat{i} + dz dx \hat{j} + dx dy \hat{k})$$

where

$$\vec{F} = x \hat{i} + y \hat{j} + z \hat{k}$$

$$\text{DIV } \vec{F} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x \hat{i} + y \hat{j} + z \hat{k})$$

$$\text{DIV } \vec{F} = 1 + 1 + 1$$

$$\text{div } \vec{F} = 3$$

By Gauss Divergence theorem

$$\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \text{div } \vec{F} dv$$

$$\iint_S x dy dz + y dz dx + z dx dy =$$

$$\iiint_V 3 dv$$

$$= \frac{1}{4} \left(\frac{n^3}{3} - 6n^2 + 36n \right)_0^6$$

$$= \frac{1}{4} \left(\frac{6^3}{3} - 6 \times 6^2 + 36 \times 6 \right)$$

$$= \frac{1}{4} \left(\frac{216}{3} - \cancel{216} + \cancel{216} \right)$$

$$= \frac{\cancel{216}^{54} 18}{\cancel{4} \times 3}$$

$$= 18$$

Q.5 Verify divergence theorem for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ taken over the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

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$$\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$$

$$\text{div } \vec{F} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (4xz\hat{i} - y^2\hat{j} + yz\hat{k})$$

$$\text{div } \vec{F} = \frac{\partial}{\partial x}(4xz) - \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(yz)$$

$$\text{div } \vec{F} = 4z - 2y + y$$

$$\boxed{\text{div } \vec{F} = 4z - y}$$

volume integral

$$\iiint_V \text{div } \vec{F} \, dv = \int_0^1 \int_0^1 \int_0^1 (4z - y) \, dz \, dy \, dx$$

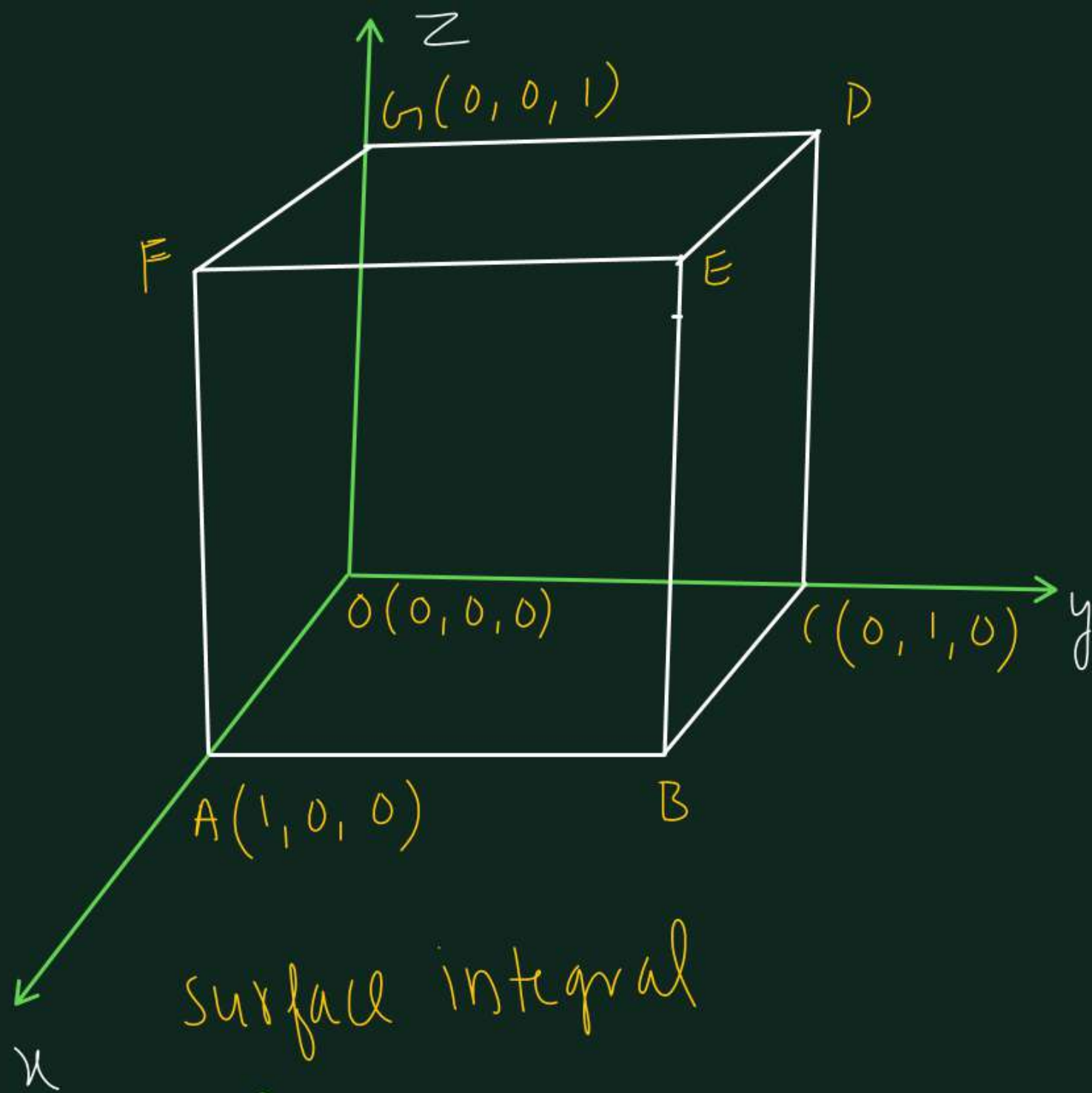
$$= \int_0^1 \int_0^1 (2z^2 - yz) \Big|_0^1 \, dy \, dx$$

$$= \int_0^1 \int_0^1 (2 - y) \, dy \, dx = \int_0^1 \left(2y - \frac{y^2}{2} \right) \Big|_0^1 \, dx$$

$$= \int_0^1 \frac{3}{2} \, dx = \frac{3}{2} (x) \Big|_0^1$$

$$= \frac{3}{2}$$

①



surface integral

$$= \iint_S \vec{F} \cdot \hat{n} \, ds$$

where, S consist 6 surfaces of cube

For OABC

$$\begin{array}{l|l} \hat{n} = -\hat{k} & dz = 0 \\ z = 0 & ds = dndy \end{array}$$

$$\iint \vec{F} \cdot \hat{n} \, ds = \iiint (4xz\hat{i} - y^2\hat{j} + yz\hat{k}) \cdot (-\hat{k}) \, dndy$$

$$= \iint (-yz) \, dndy$$

$$= 0$$

For DEFG

$$\hat{n} = \hat{k}$$

$$dz = 0$$

$$z = 1$$

$$ds = dndy$$

$$\iint \vec{F} \cdot \hat{n} ds = \int \int (4nz\hat{i} - y^2\hat{j} + yz\hat{k}) \cdot (\hat{k}) dndy$$

$$= \int \int yz dndy = \int_0^1 \int_0^1 y dy dn$$

$$= \int_0^1 \left(\frac{y^2}{2} \right)_0^1 dn = \frac{1}{2} \int_0^1 dn$$

$$= \frac{1}{2}$$

For BCDE

$$\hat{n} = \hat{j}$$

$$dy = 0$$

$$y = 1$$

$$ds = dndz$$

$$\iint \vec{F} \cdot \hat{n} ds = \int_0^1 \int_0^1 (4nz\hat{i} - y^2\hat{j} + yz\hat{k}) \cdot (\hat{j}) dndz$$

$$= \int_0^1 \int_0^1 -y^2 dndz = - \int_0^1 \int_0^1 dz dn$$

$$= - \int_0^1 (z)_0^1 dn = - \int_0^1 dn$$

$$= -1$$

For OCU

$$\hat{n} = -\hat{i} \quad dn = 0$$

$$n = 0 \quad ds = dy dz$$

$$\iint \vec{F} \cdot \hat{n} \, ds = \int_0^1 \int_0^1 (4yz \hat{i} - y^2 \hat{j} + yz \hat{k}) (-\hat{i}) \, dy \, dz$$

$$= \int_0^1 \int_0^1 (-4yz) \, dy \, dz$$

$$= \int_0^1 \int_0^1 0 \, dy \, dz = 0$$

Hence

$$\iint_S \vec{F} \cdot \hat{n} \, ds = 0 + \frac{1}{2} - 1 + 0 + 2 + 0$$

$$\boxed{\iint \vec{F} \cdot \hat{n} \, ds = \frac{3}{2}} \quad \text{--- (2)}$$

From (1) and (2)

$$\boxed{\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \text{div} \vec{F} \, dv}$$

Hence divergence theorem is verified

P.Q.3 Verify divergence theorem for $\vec{F} = x^2\hat{i} + z\hat{j} + yz\hat{k}$ taken

over the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

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P.Q.4 Verify divergence theorem for $\vec{F} = (x^3 - yz)\hat{i} + (y^3 - zx)\hat{j} + (z^3 - xy)\hat{k}$ taken over the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

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P.Q.5 Evaluate $\iint (a^2x^2 + b^2y^2 + c^2z^2)^{-\frac{1}{2}}, d\bar{S}$ where S is the surface of the
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ellipsoid $ax^2 + by^2 + cz^2 = 1$.

Ans. $\frac{4\pi}{\sqrt{abc}}$

Syllabus Completed

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Thank You